

The other story told by Urban Ecology in the Global South

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Alexander von
HUMBOLDT
STIFTUNG

Bielefeld 2025



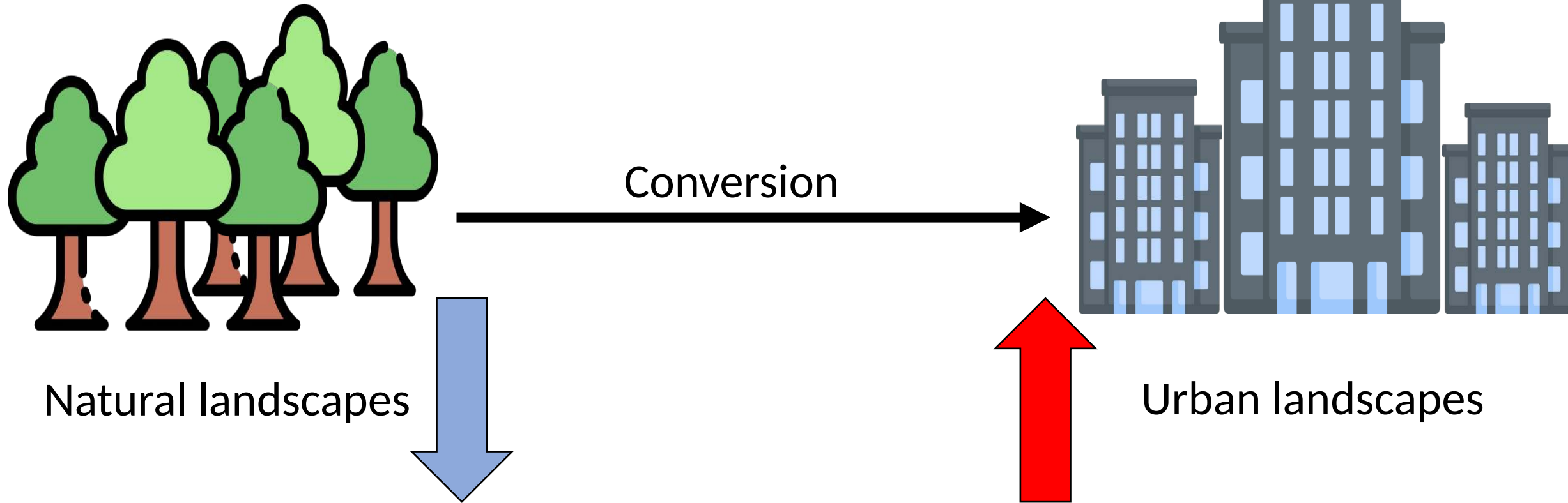
iDiv

German Centre for Integrative
Biodiversity Research (iDiv)
Halle-Jena-Leipzig

Outline

- Brief intro on why urban ecological relationships should be different in the Tropics and developing countries
- Recent results on the effects of urbanisation on phenotypes using frogs as study organisms
- Population level study
- Community scale
- Closing remarks

Formation of novel *urban ecosystems*



Consequences of novel *urban ecosystems*

a. Habitat modification



b. Connectivity



c. Heterogeneity



d. Novel disturbances



e. Biotic interactions

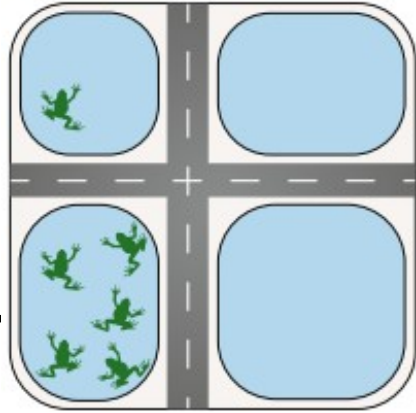


f. Scale

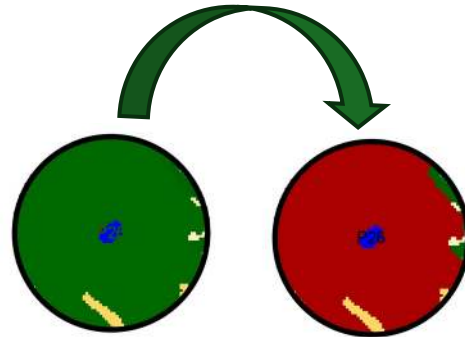


Introduction

Fragmentation



Land use changes



Heat islands

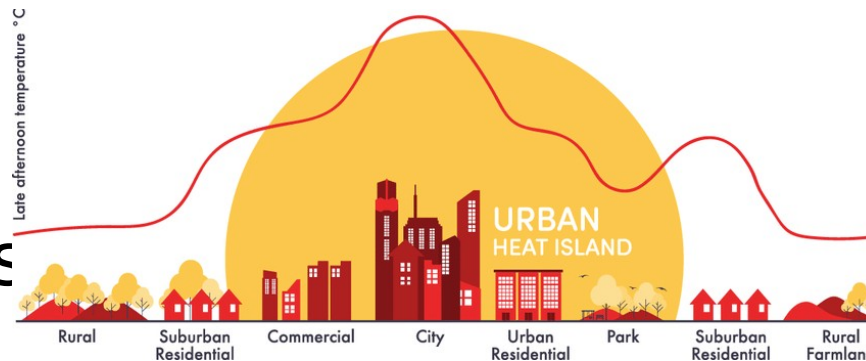


Fig. Kamyar Fuladlu

Urbanization negatively impacts frog diversity at continental, regional, and local scales

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Alistair G.B. Poore^b, Jodi J.L. Rowley^{a,d}

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^bEcology & Evolution
NSW, Australia

^cCzech University of

^dAustralian Museum

Biological Journal of the Linnean Society, 2022, XX, 1–15. With 6 figures.

75,

The country toad and the city toad: comparing morphology of invasive cane toads (*Rhinella marina*) from rural and urban environments

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Received 21 May 2022; revised 1 June 2022; accepted 1 June 2022

Urbanization is a principal driver of biodiversity loss. Urban environments are hotspots for native species responding to urbanization. We investigated the response of invasive cane toads (*Rhinella marina*) to urbanization using morphological data. We found that the morphological traits of cane toads in urban environments were significantly different from those in rural environments.

Variation in age, body size, and reproductive traits among urban and rural amphibian populations

Matthew A. Jennette^{1,2} · Joel W. Snodgrass^{1,3} · Don C. Forester¹

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Abstract

Although amphibians use human-created habitats in urban landscapes, few studies have investigated the quality of these habitats. To assess habitat quality of stormwater management ponds and adjacent urban uplands for wood frogs (*Lithobates sylvaticus*) and American toads (*Anaxyrus americanus*), we compared life history characteristics between populations breeding across an urbanization gradient. Specifically, we compared body size, ages of breeding adults, and female reproductive investment among urban, suburban, and rural populations in Baltimore County, Maryland, USA. Although there was variation in age at maturity among populations, ages of breeding adults did not differ among urban, suburban, and rural areas. Maternal body size strongly influenced reproductive investment in both species, but relationships did not vary among urban, suburban, and rural populations. Adult wood frogs and American toads from more urbanized landscapes were significantly smaller at age than conspecifics from rural landscapes; the magnitude of differences was similar across adult age classes. Our results suggest that in the urban and rural landscapes that we studied, adult habitats are similar in quality, but either larval or juvenile habitats may be of lower quality in urban areas.

Keywords Habitat quality · Human-created habitats · Stormwater ponds · Anurans · Wood frog · American toad · Life history



Lambert and Donihue

Habitats **changed** in the urban ecosystem

Urbanisation



Isolated

No/less vegetation

Water deficit

Silted

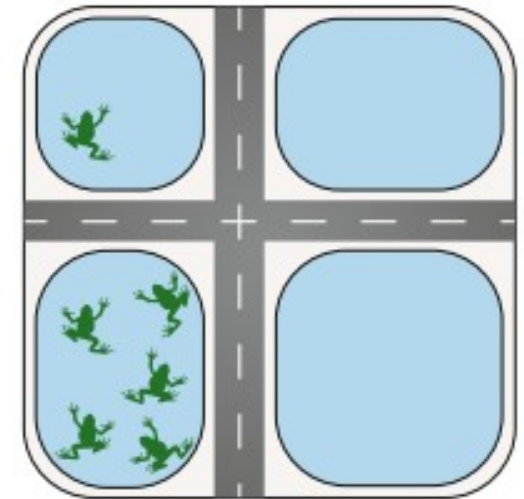
Polluted/Sewage



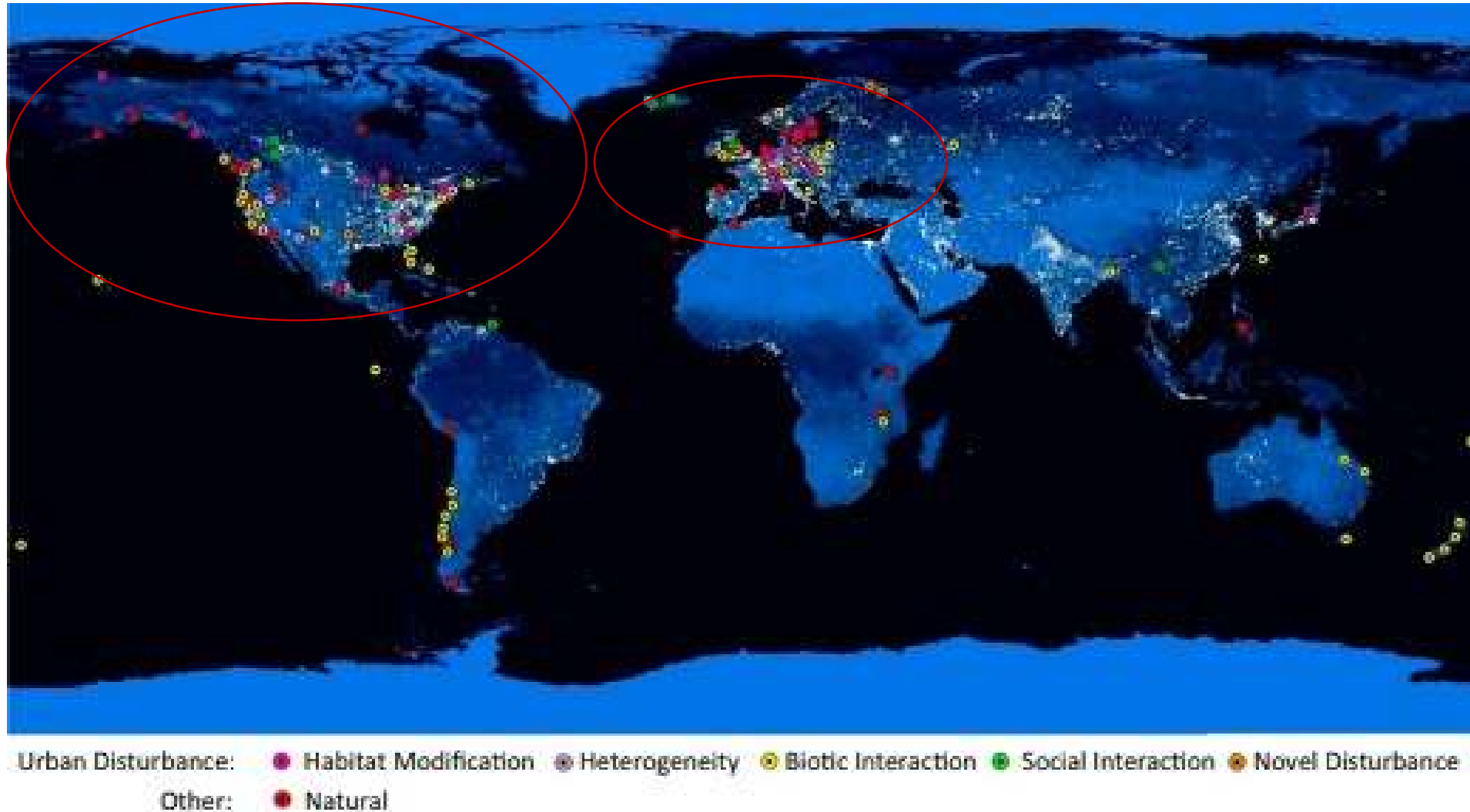
Rural pond



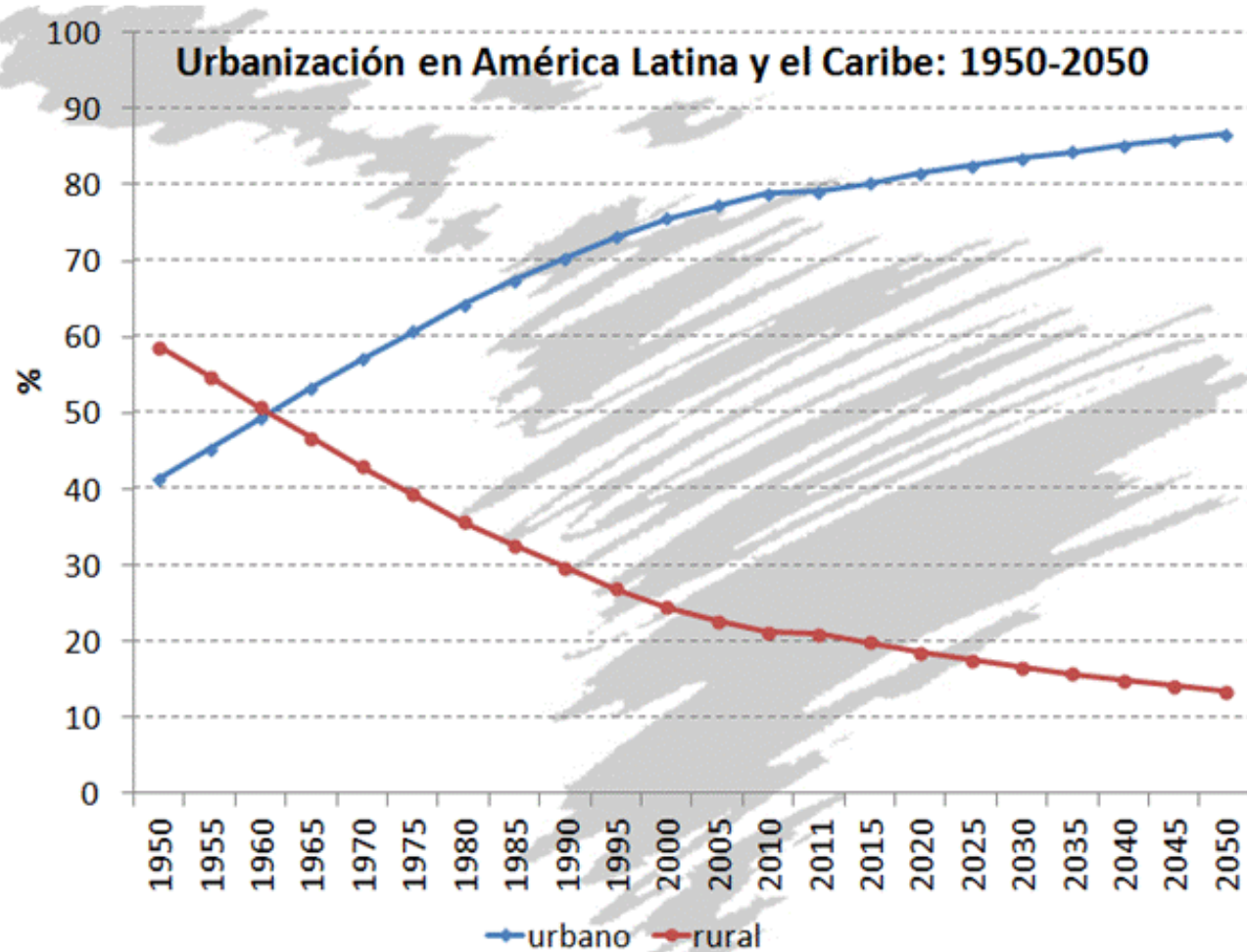
Urban pond



Majority of studies on urban ecology were done in the **Global North**

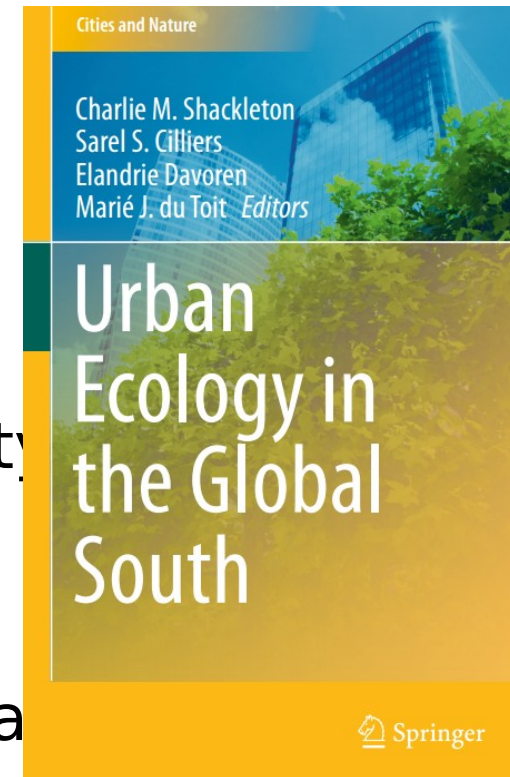


More recent, although intense urbanisation in the **Global South**



Why Ecology is different in Tropical urban ecosystems?

- Recent urbanisation...
 - Time for local adaptation/population divergence
- ...But at a fast pace
 - Strong natural selection -> Fixation of genes/phenotype
- Socio-economic dynamics
 - Less aquatic habitats remain in more urbanised areas
 - E.g., Mosquito-borne diseases (ecosystem 'disservices')
 - Distribution of and access to urban greenspaces
 - Safety concerns



Contrasting inequality in human exposure to greenspace between cities of Global North and Global South

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Accepted: 19 July 2022

Published online: 08 August 2022

Bin Chen^{1,2,3}✉, Shengbiao Wu¹, Yimeng Song⁴, Chris Webster^{2,3,5},
Bing Xu⁶✉ & Peng Gong^{2,7}✉

The United Nations specified the need for “providing universal access to

- *Global South cities experience only 1/3 of the greenspace exposure level of Global North ones;*
- *Greenspace exposure inequality (Gini: 0.47) in Global South cities is nearly double that of Global North ones (Gini: 0.27).*

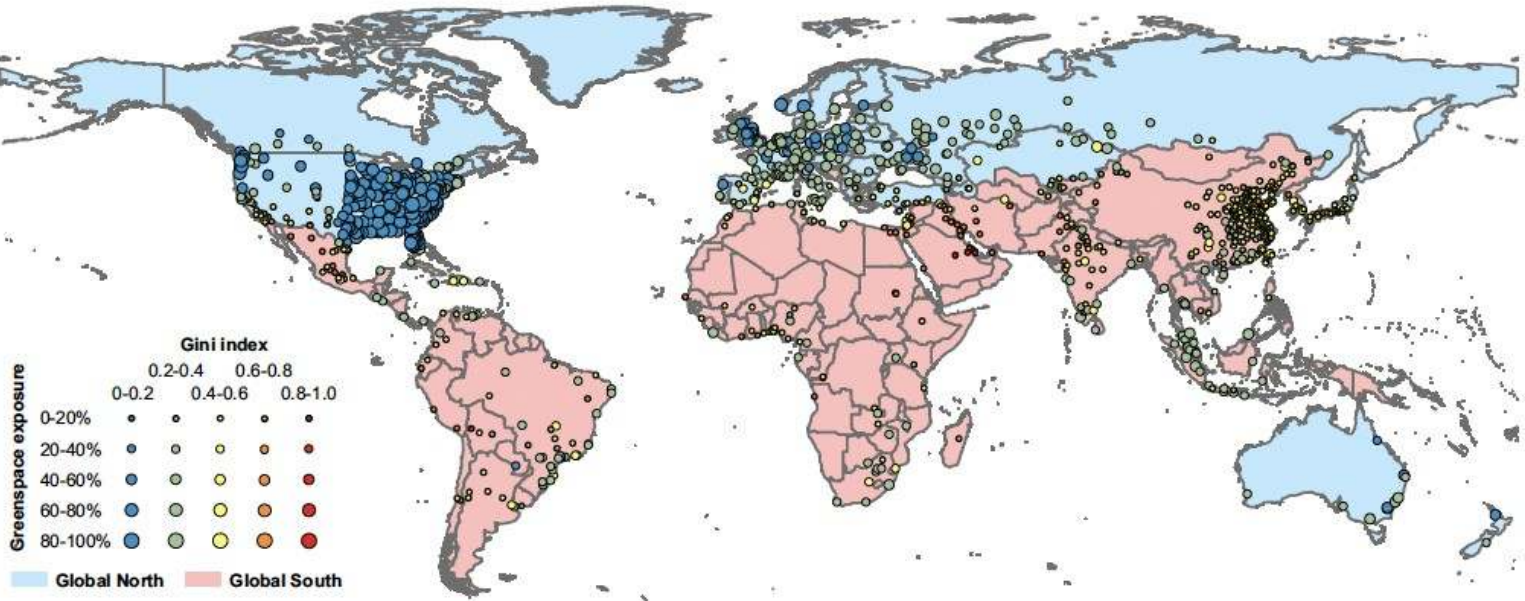


Fig. 2 | Greenspace exposure levels and the associated greenspace exposure inequalities measured by the Gini index for 1028 cities globally. Larger bubble sizes represent higher levels of greenspace exposure, and warmer colors represent

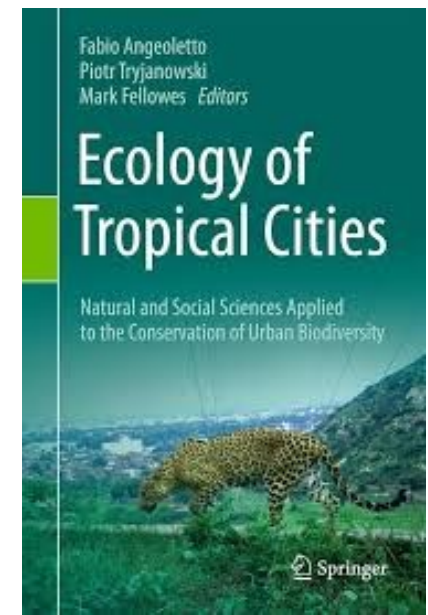
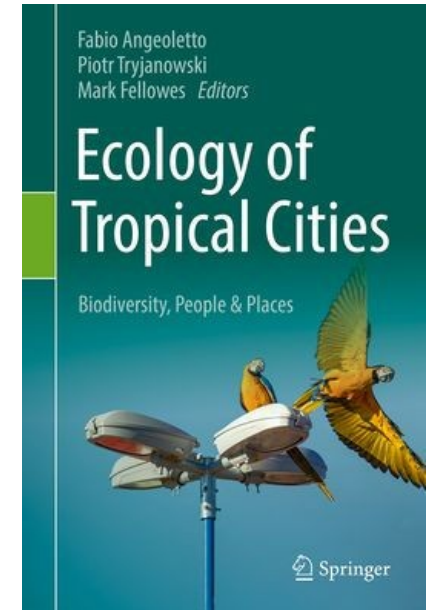
higher levels of greenspace exposure inequality. Administrative boundaries with light blue (pink) shading represent Global North (South) countries.

Table 1 | Statistics of city-level human exposure to greenspace and greenspace exposure inequality across regions

Region (# of cities)	Greenspace exposure	Gini of greenspace exposure
Global North (522)	45.84 ± 20.71%	0.24 ± 0.13
Global South (506)	14.39 ± 9.57%	0.47 ± 0.13
North America (293)	53.45 ± 22.09%	0.21 ± 0.14
South America (60)	19.89 ± 8.09%	0.41 ± 0.12
Europe (180)	39.25 ± 11.43%	0.26 ± 0.08
Africa (63)	17.66 ± 10.21%	0.46 ± 0.14
Asia (420)	13.49 ± 9.52%	0.47 ± 0.12
Australia/Oceania (12)	42.51 ± 8.33%	0.22 ± 0.06
Global (1028)	30.36 ± 22.58%	0.35 ± 0.17

Why Ecology is different in Tropical urban ecosystems?

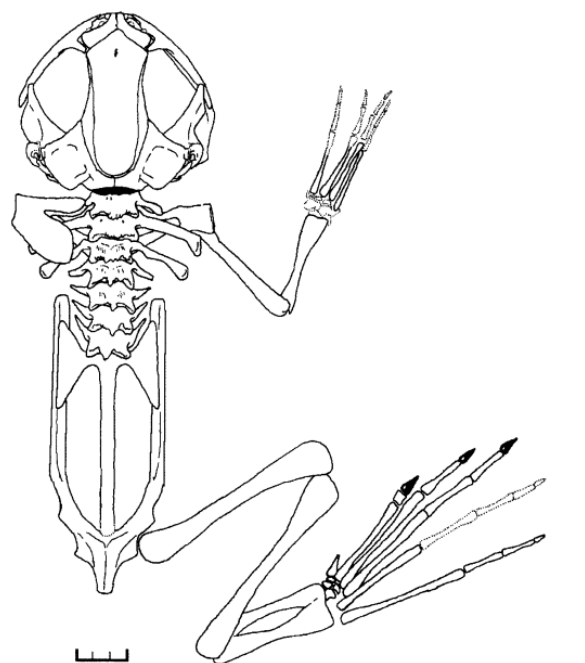
- Yet, Tropical areas are the richest in biodiversity
 - Urban ecosystems can be oasis and provide opportunities for conservation
- Threats other than urbanisation
- SDGs UN 2030 objectives



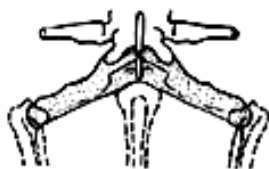
Two studies on the effects of urbanisation on
quantitative phenotypic traits of aquatic
organisms

Why study frogs?...

...Because they're cool



Necturus



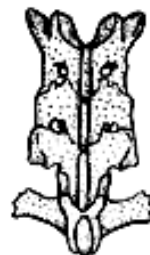
Frog



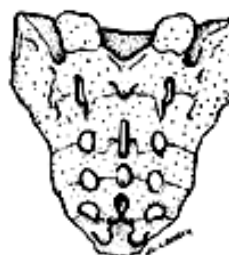
Alligator



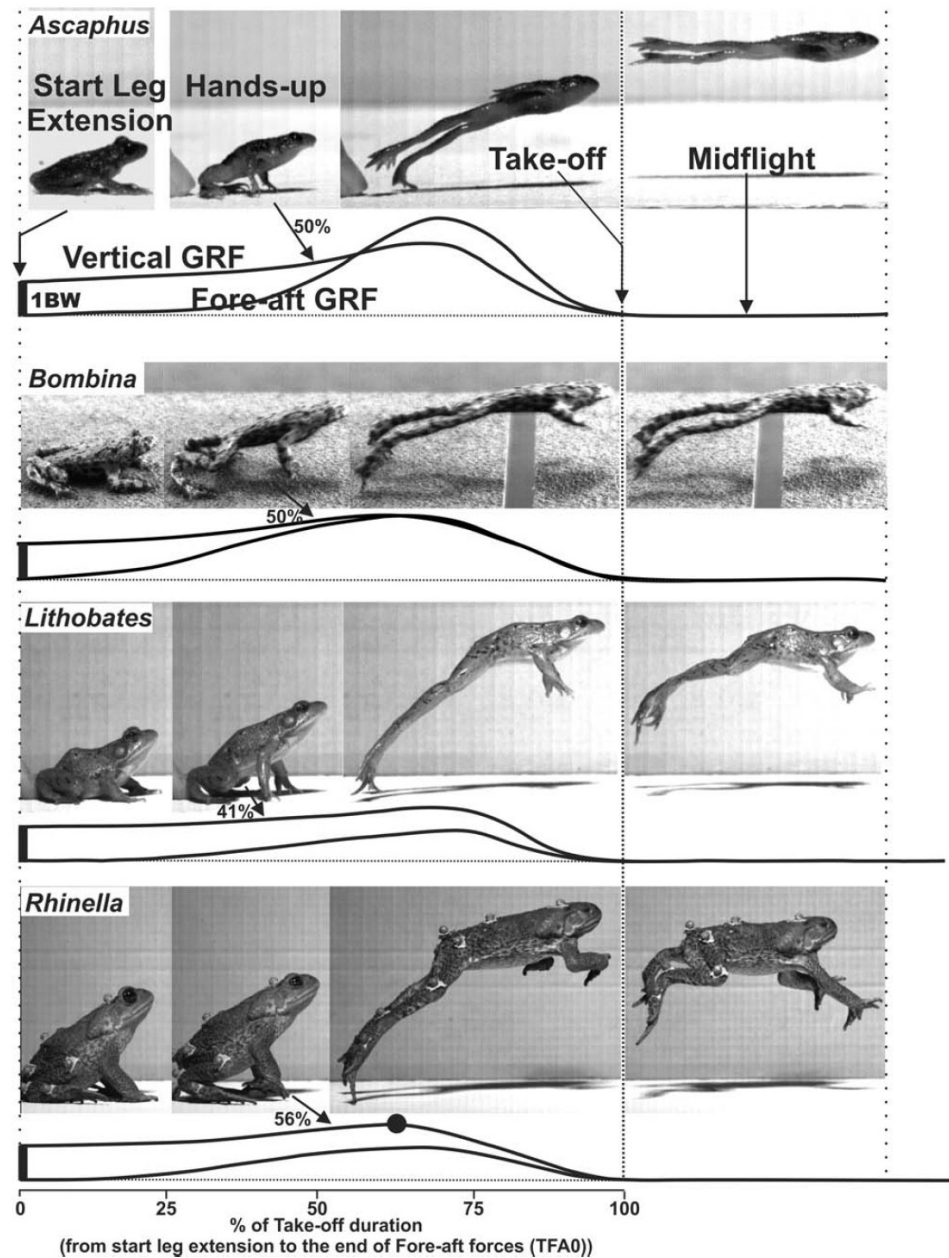
Cat

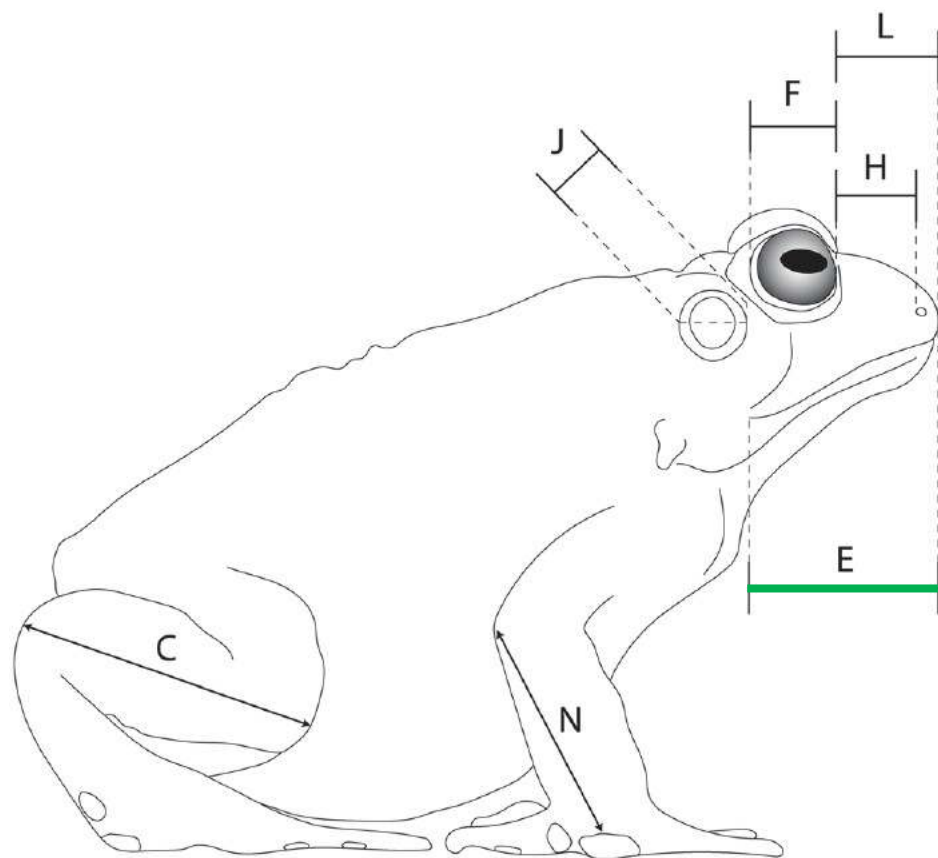


Hamster



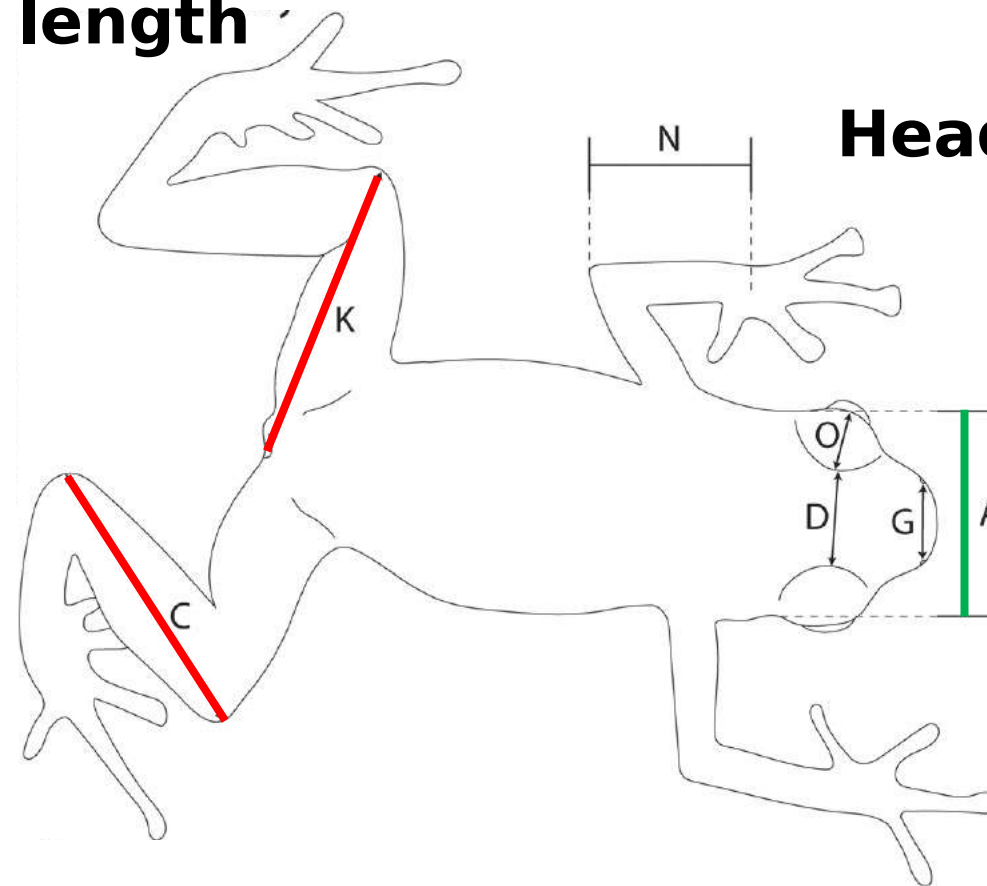
Man





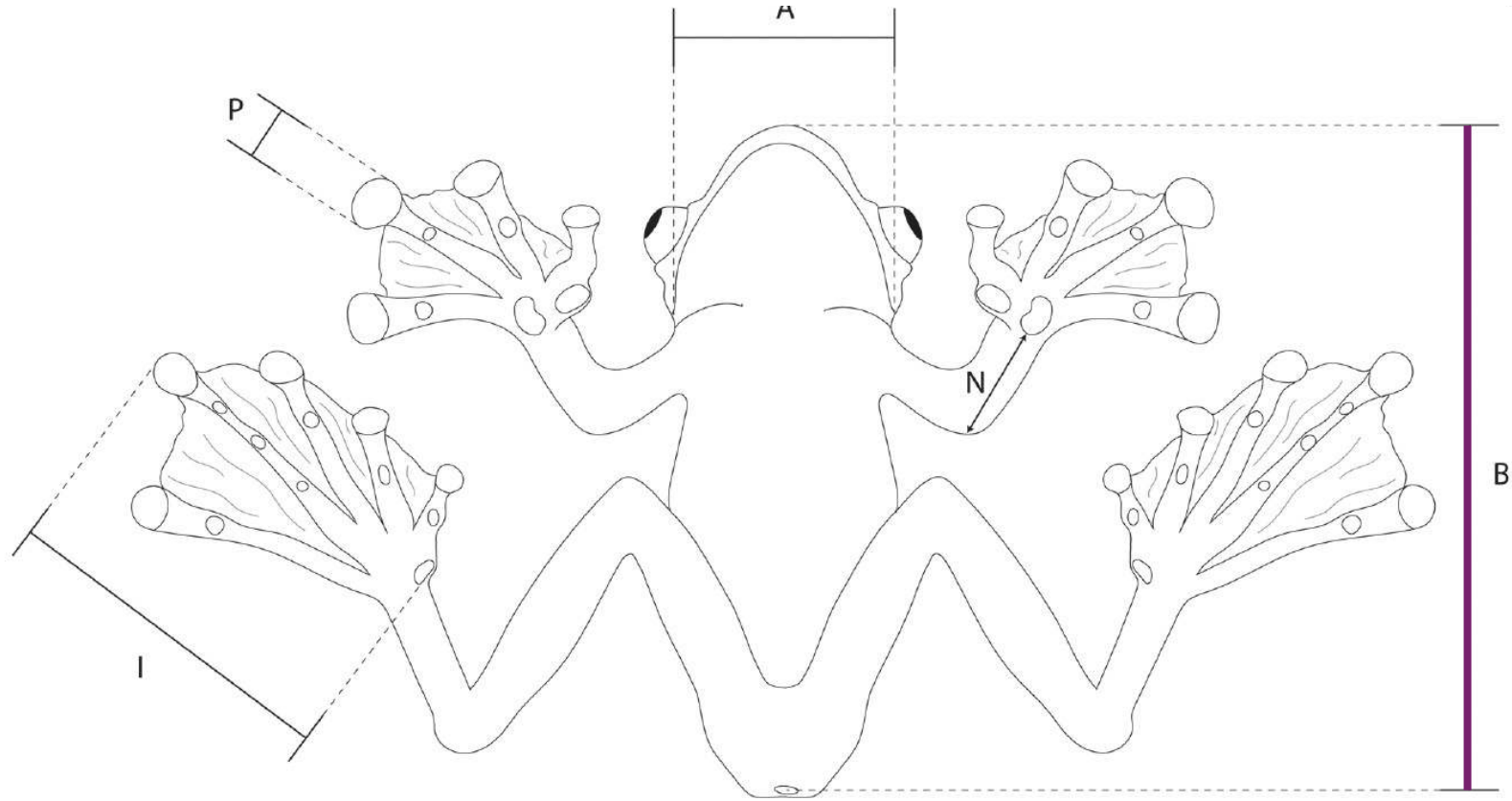
Jaw/head length

Relative leg length

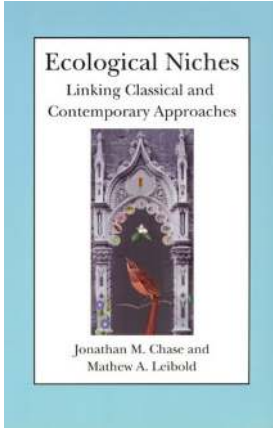


Head width

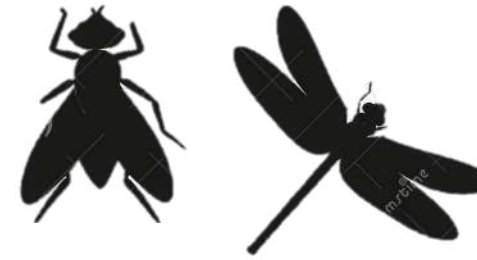
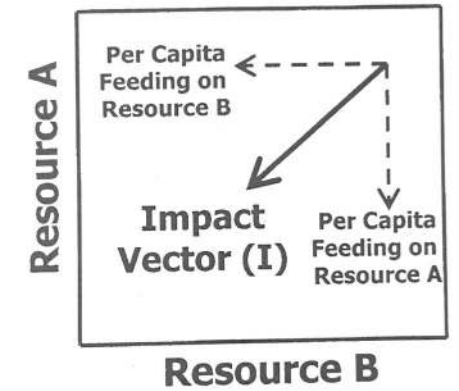
Body size (Snout-vent length)



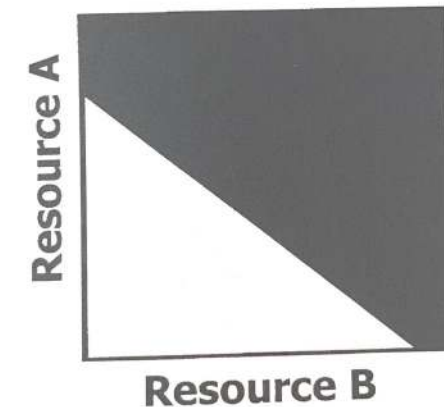
Why measuring those specific traits?



Impact Niche (diet)



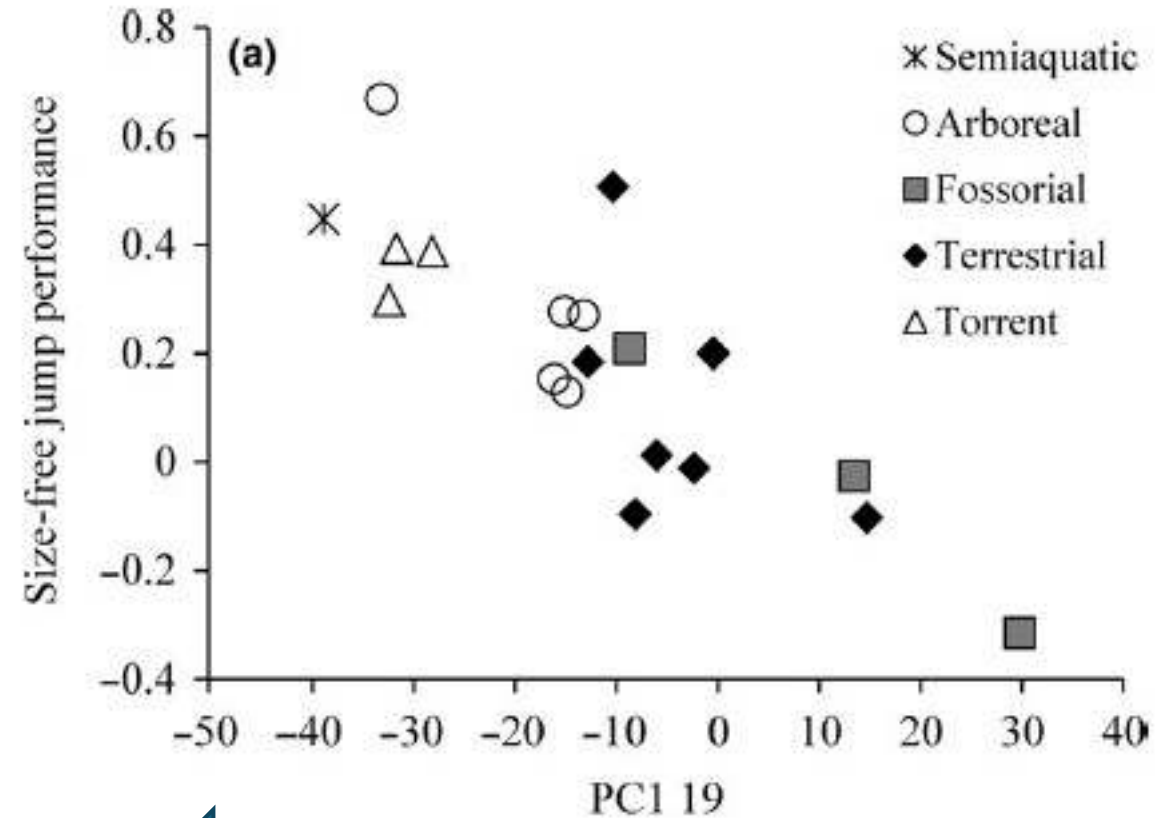
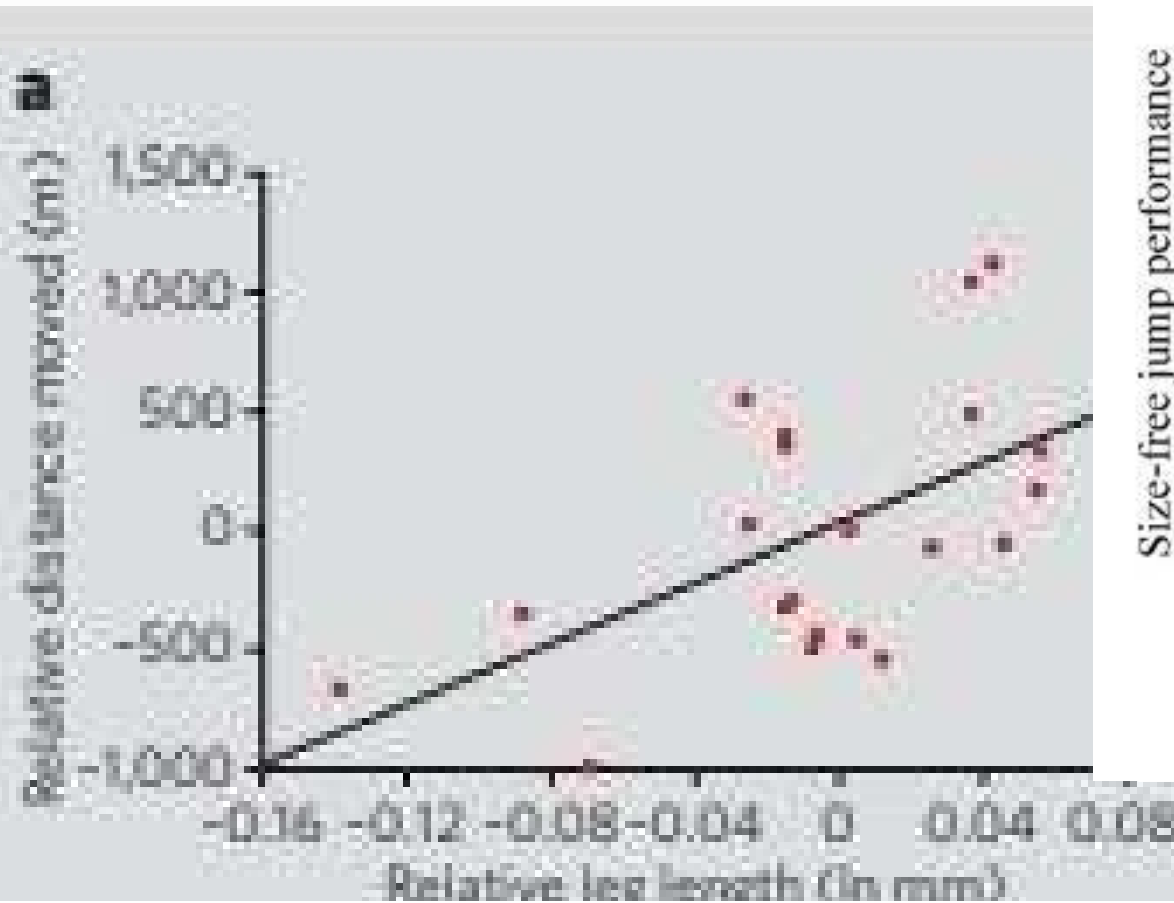
Requirement Niche (Habitat use)



Functional basis

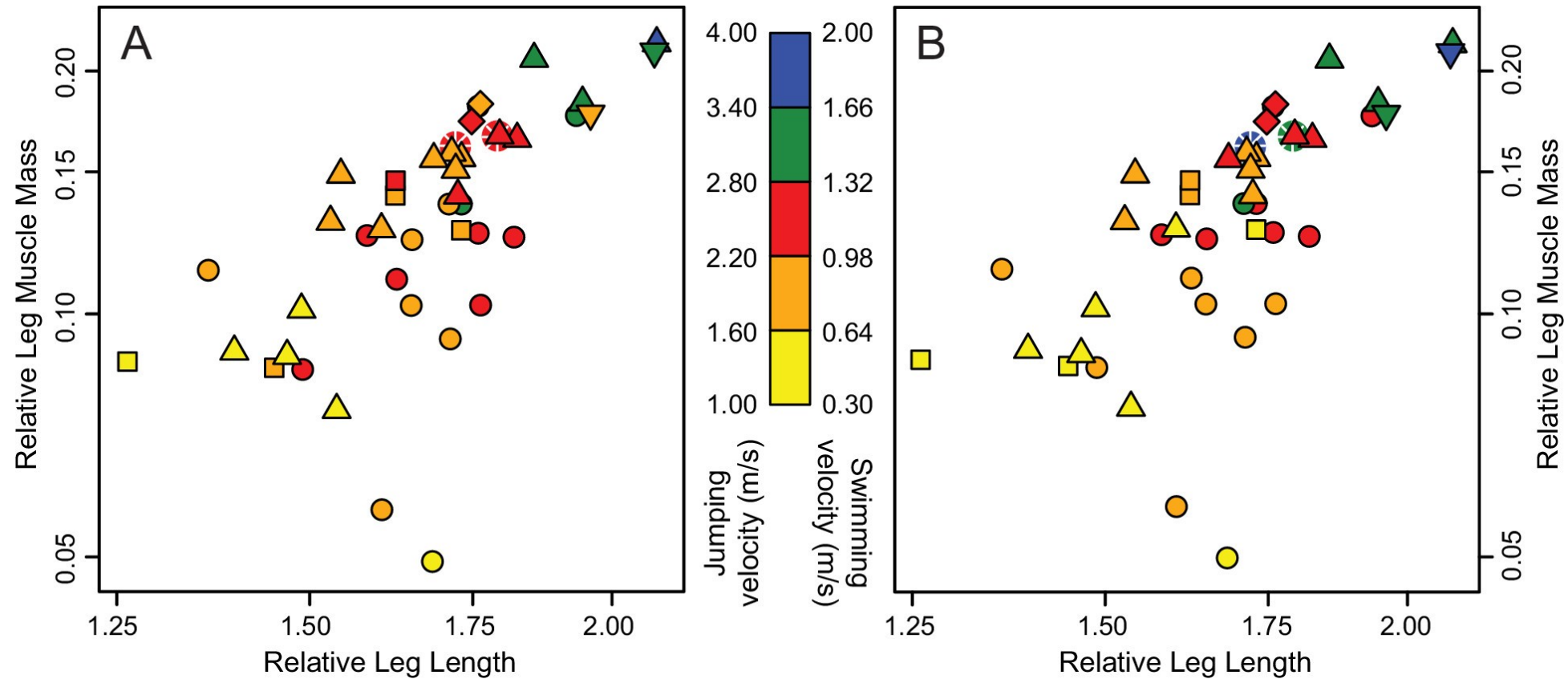
- Link between form and function -

Relative leg length is positively related to distance moved and jumping performance



Fore and hind limbs length

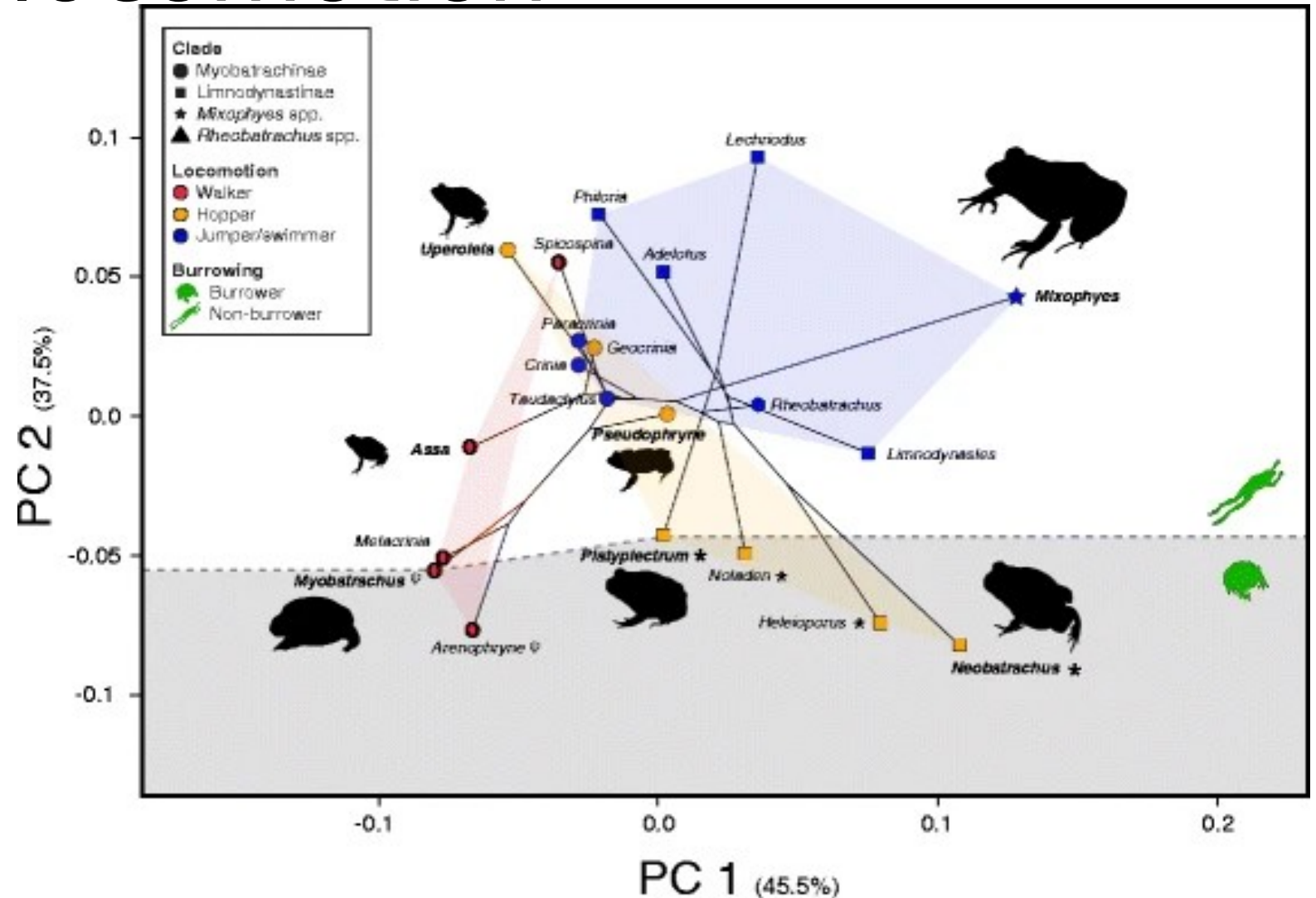
Relative leg length influences jumping and swimming velocity



Limb shape is related to habitat use and mode of locomotion

Myobatrachidae

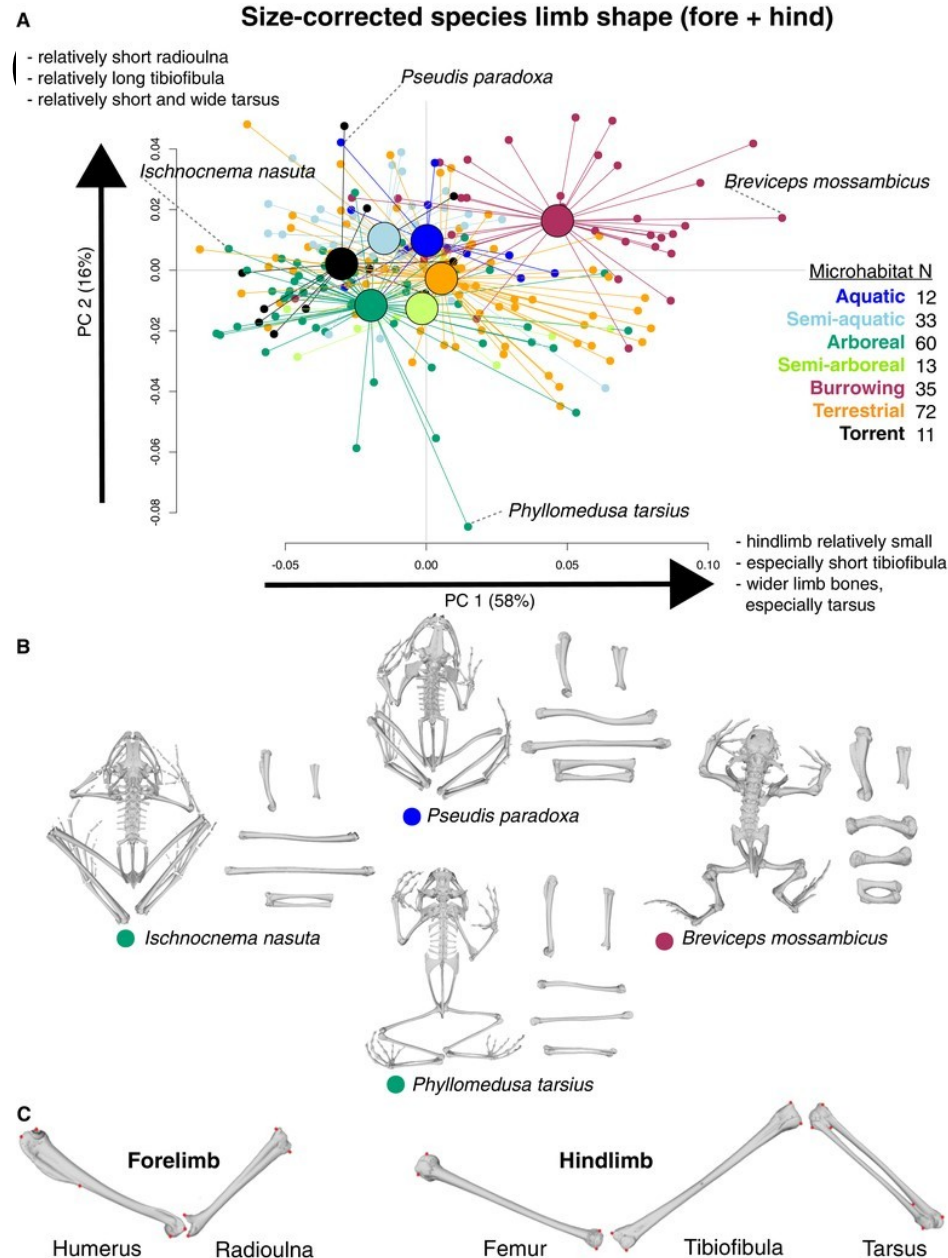
Australia



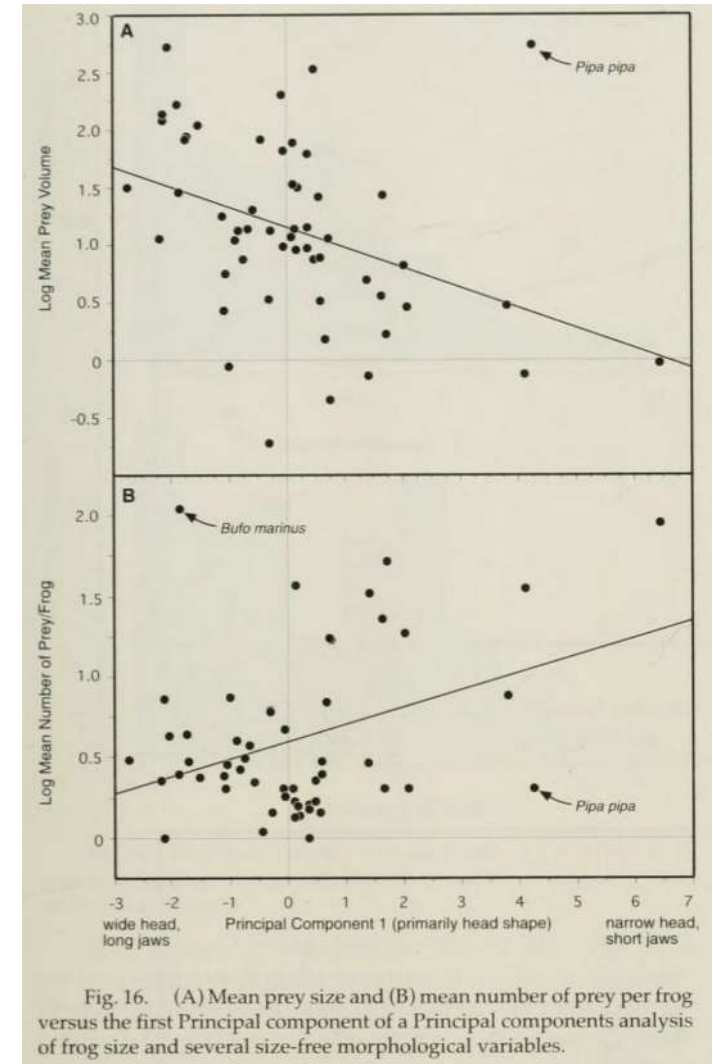
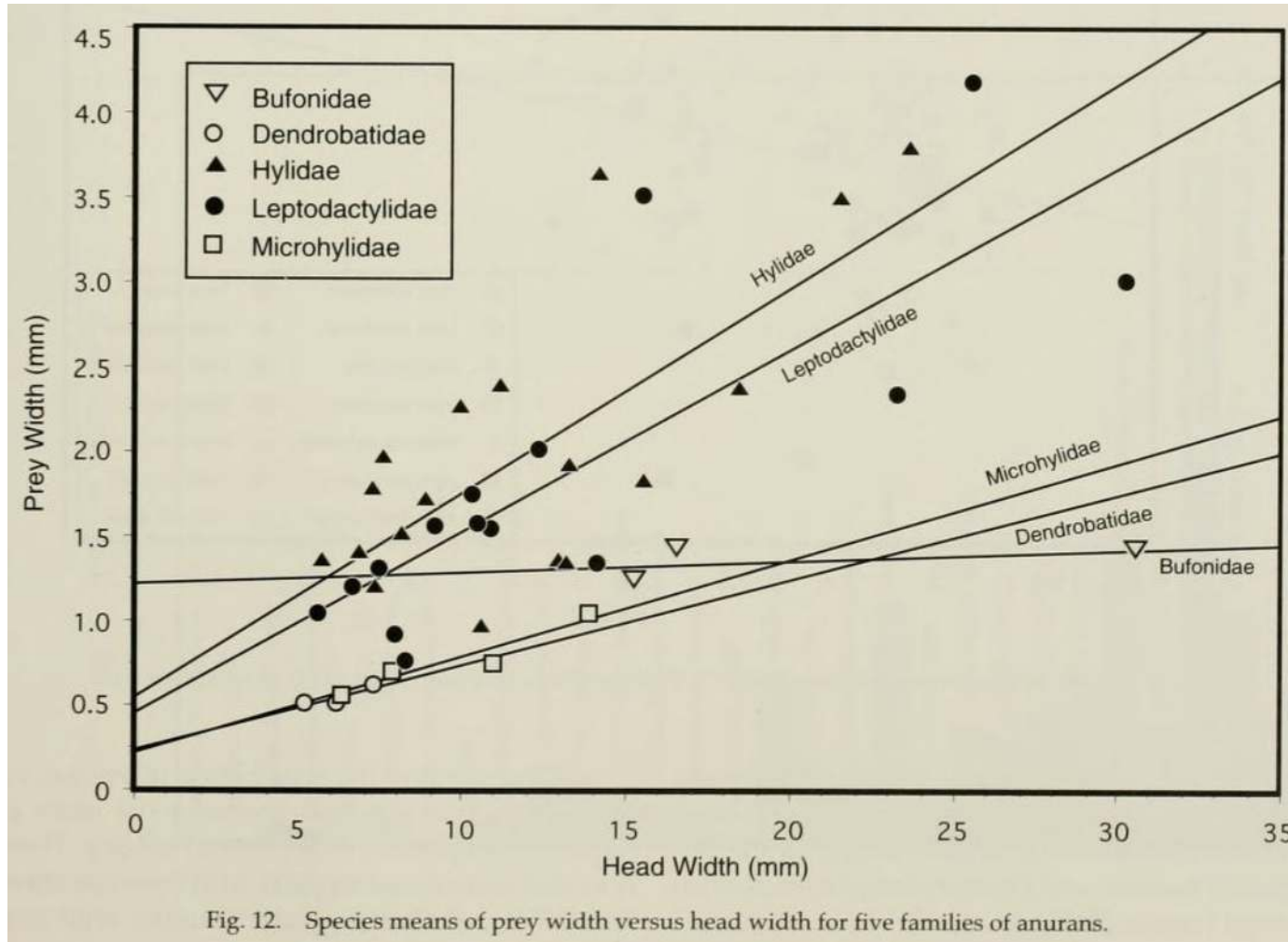
Limb shape is related to habitat use and mode of loc

Larger
samplin
g
coverag
e

Stepanova & Womack 2020 Evolution



Head width and jaw length influence the size, amount, and volume of prey consumed



Jaw length also influences the type of prey consumed

SKULL SHAPE IN FROGS—CORRELATIONS WITH DIET

SHARON B. EMERSON

- Frogs that eat relatively **small, slow prey** have relatively **short jaws** and a symmetrical feeding cycle where equal amounts of time are spent catching and bringing prey into the mouth;
- frogs that eat relatively **large, slow prey** have relatively **long jaws** and an asymmetrical feeding cycle where time to prey capture is less than the time to bring prey into the mouth;
- snail-eating frogs have **lower jaws** with a relatively **long distance from the insertion point of the adductor muscle mass to the jaw articulation**.

Population level effects of urbanisation on phenotypic variation



**Marcos R
Severgnini (PhD)**

WILEY	ECOLOGY
The effect of urbanisation and local environmental heterogeneity on phenotypic variability of a tropical treefrog	
Journal:	<i>Ecology</i>
Manuscript ID	ECY25-0086
Wiley - Manuscript type:	Report
Date Submitted by the Author:	22-Jan-2025

**In
review**

Phenotypic traits

Linear measurements of **761** individuals

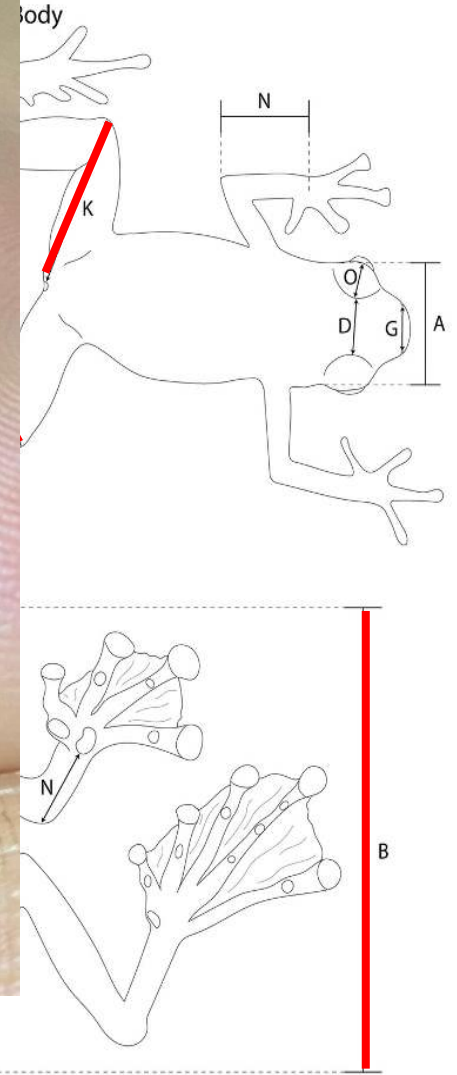


Dendropsophus
nanus

Measurements:

Mean

Coefficient of variation

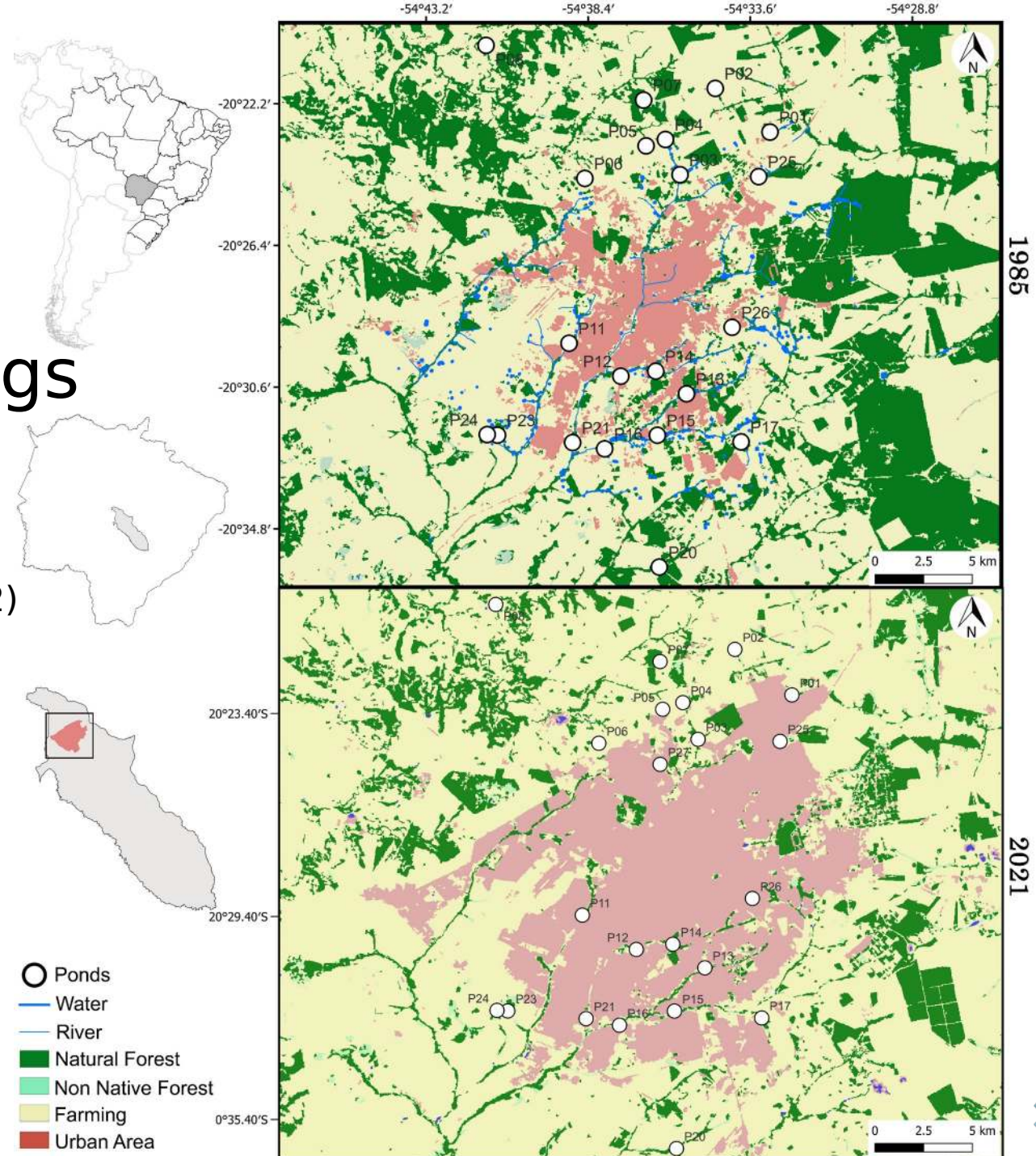


Methods

Tropical small treefrogs
live
~ 12 years

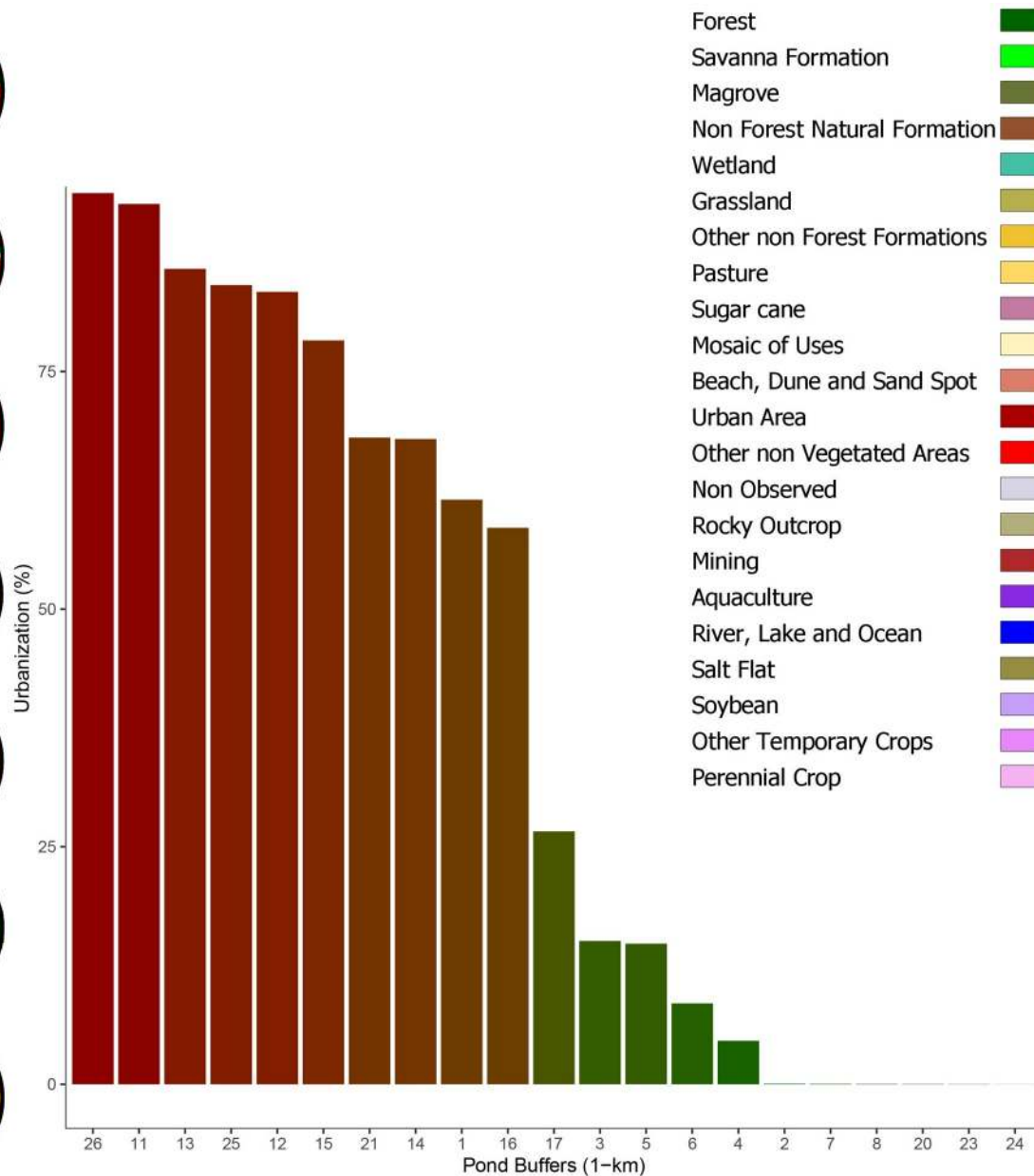
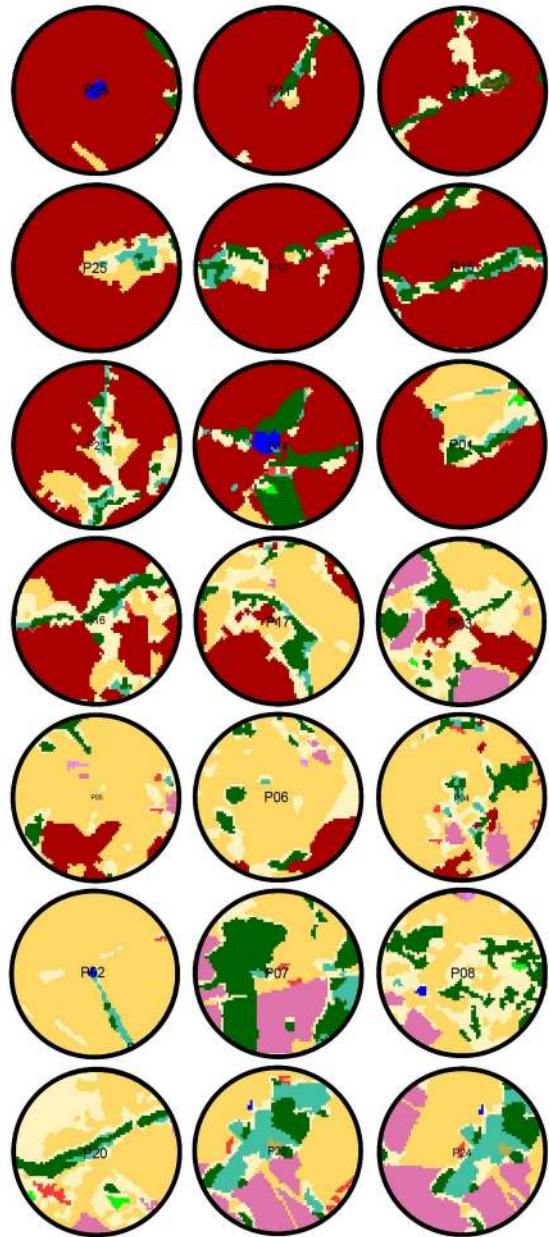
(Stark & Meiri 2018, Reinke et al. 2022)

At least 2-3 generations

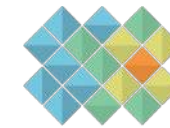


36 yr

Sampling sites



21 ponds along an urbanization gradient (from 0% to 94% impervious surface) – 1-km buffer



MAPBIOMAS
[BRASIL]



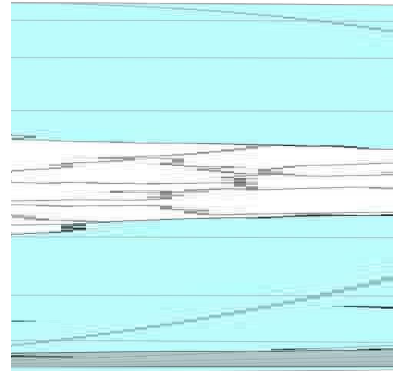
Urbanisation rate

Space-for-time substitution

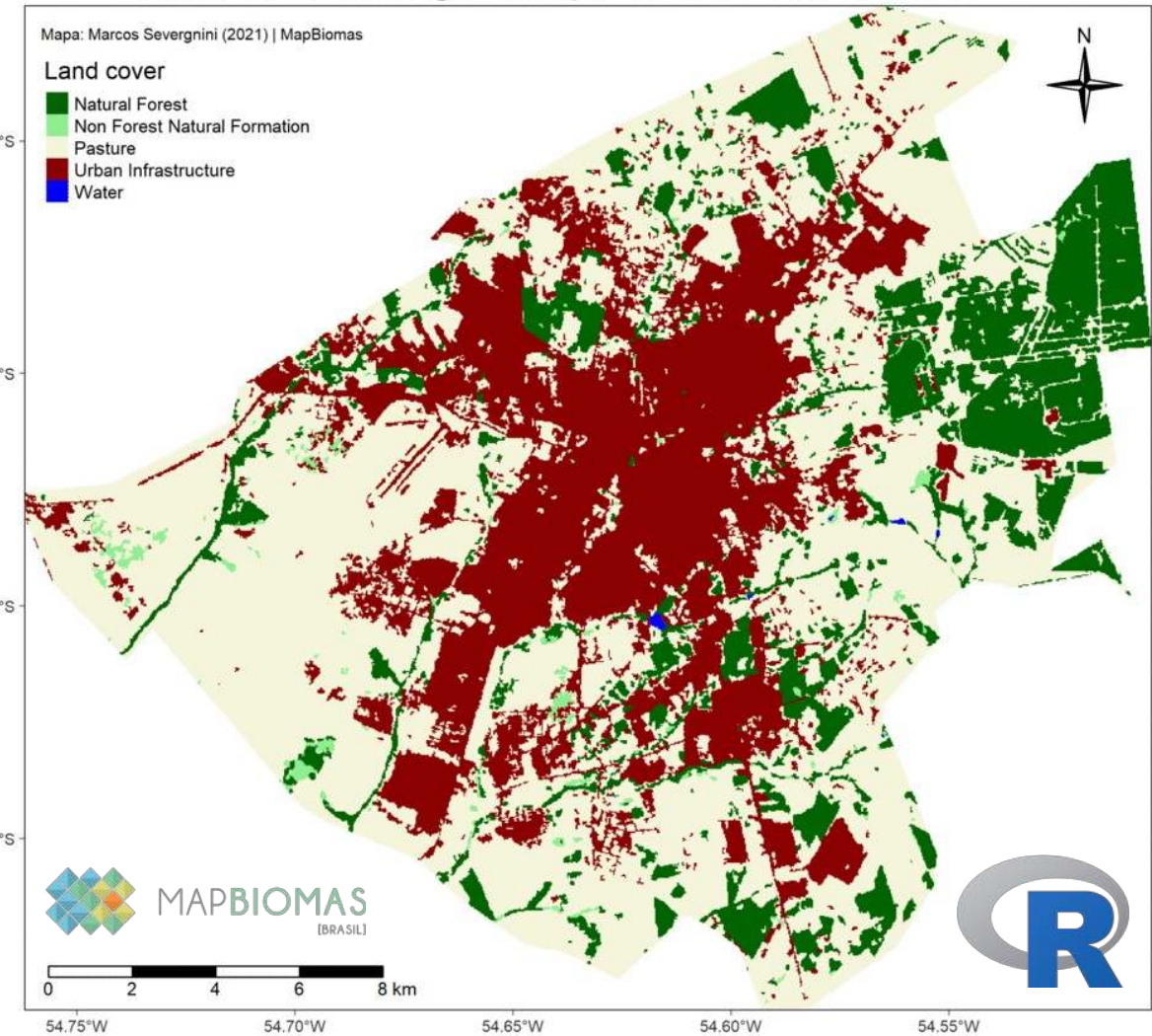
Wogan and Wang 2018

Urbanisation rate:

(% urbanisation **2021** - % **1985**)
36 years



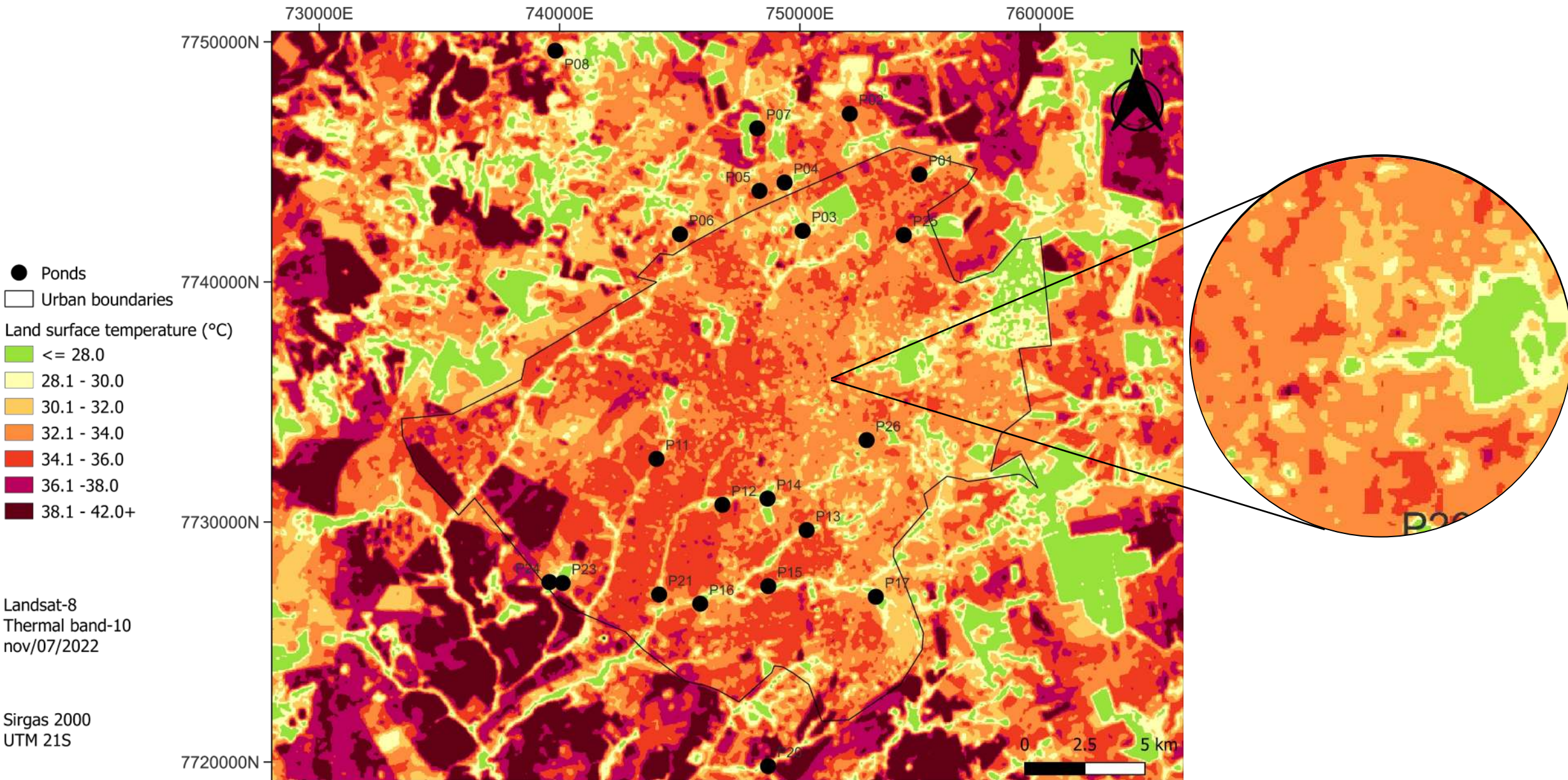
Land cover change Campo Grande 1985, Brazil



Urban Heat Island

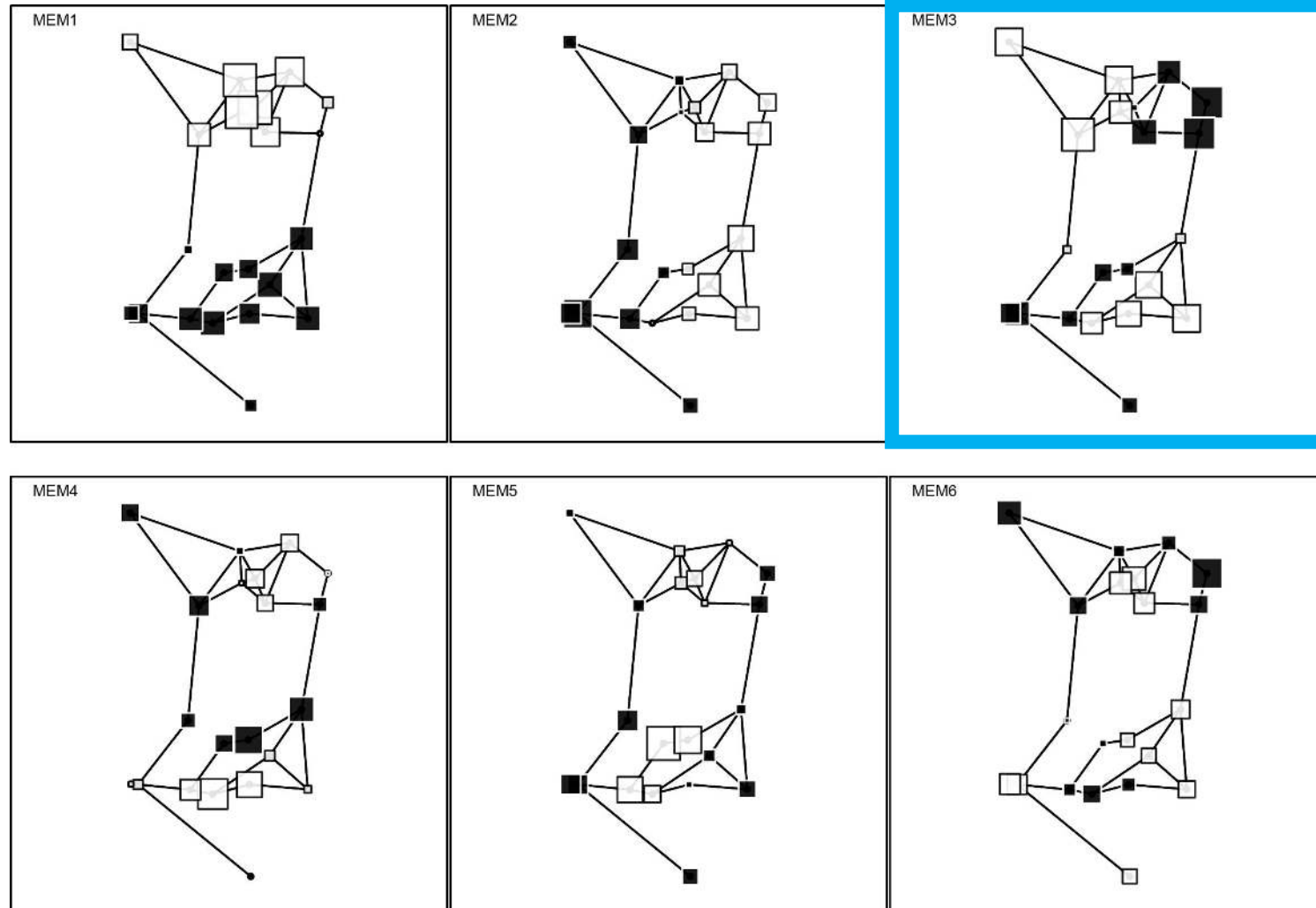
Marcos Severgnini

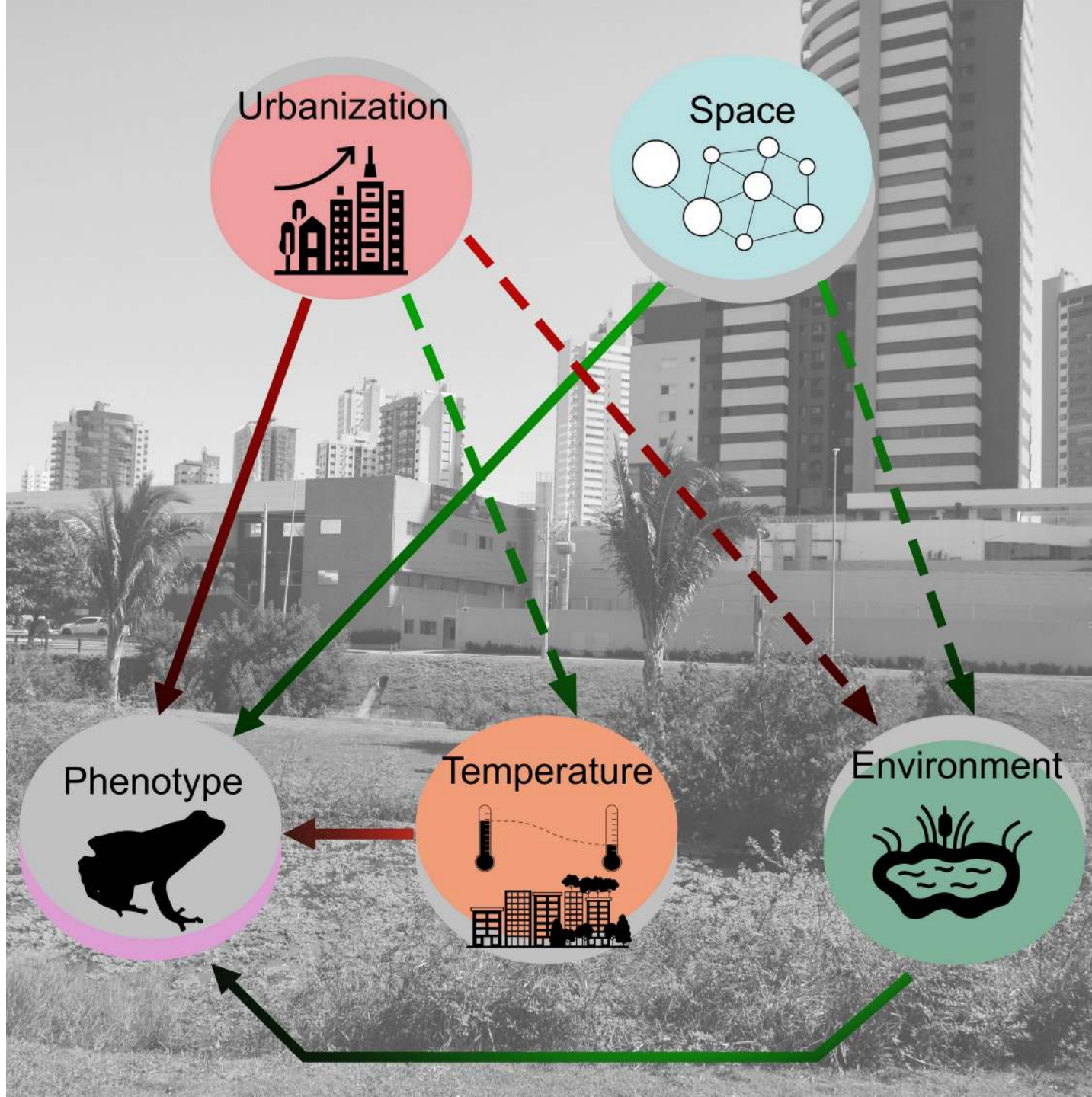
Land surface temperature of Campo Grande - Urban Heat Islands

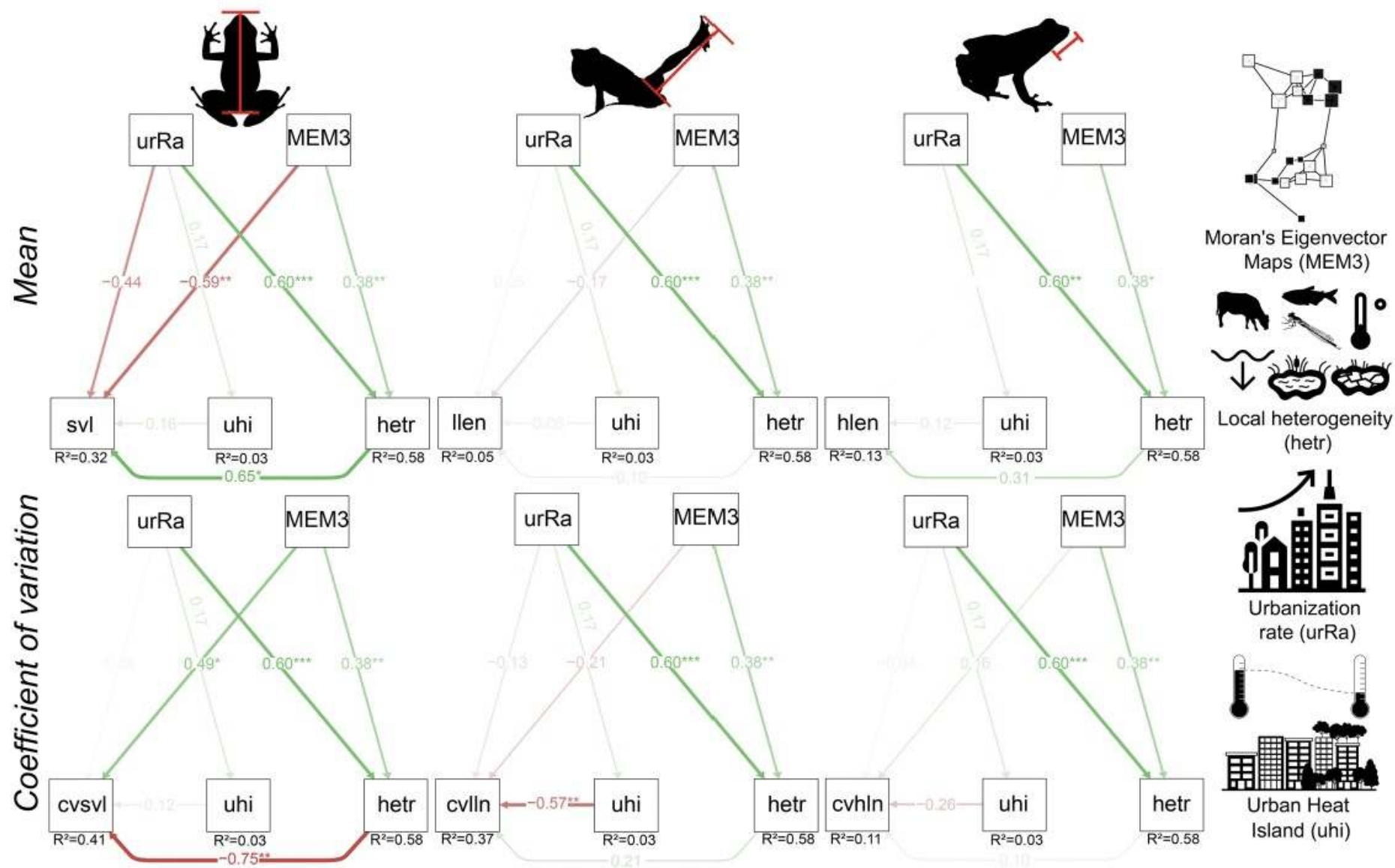


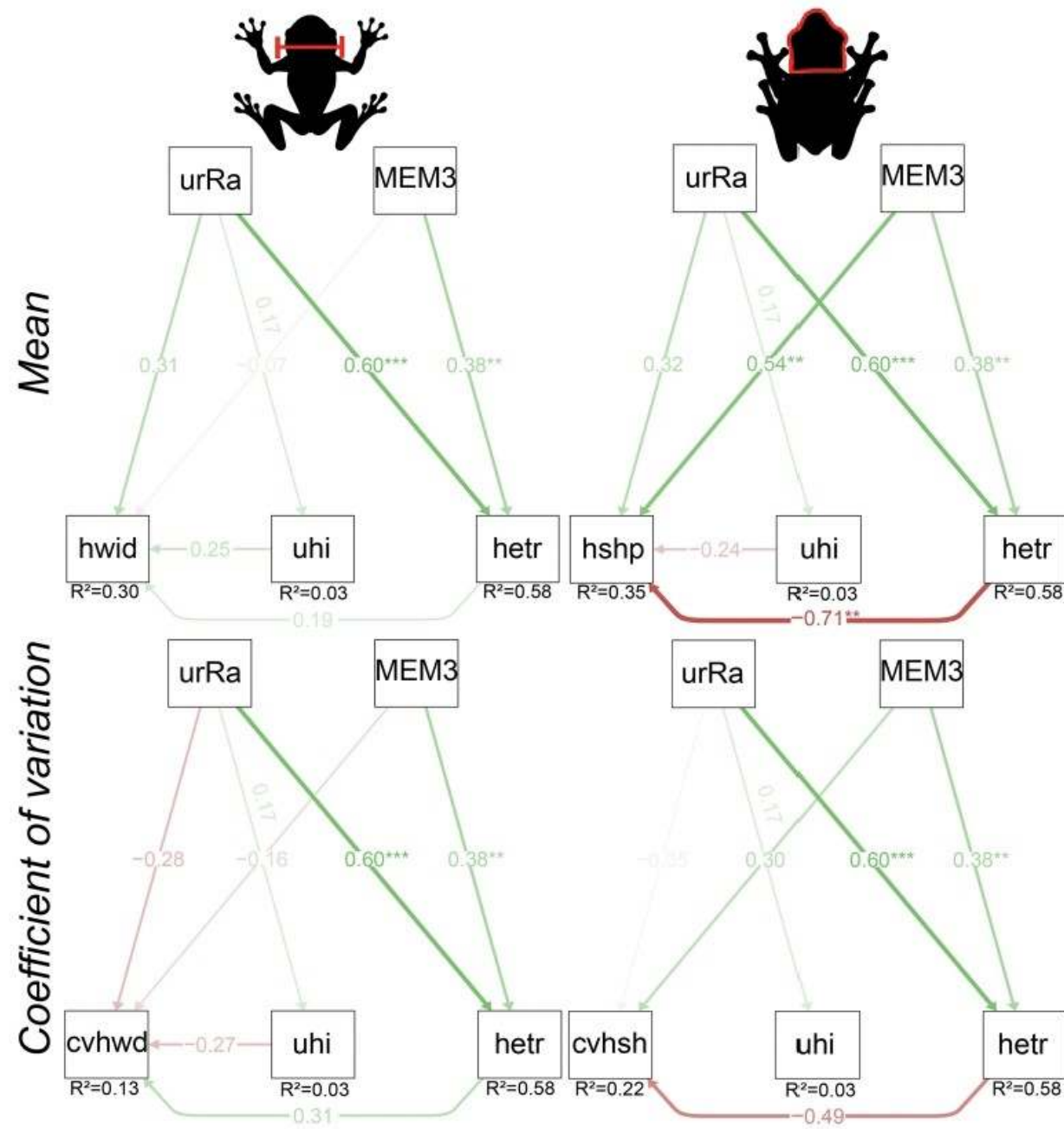
Methods

Spatial arrangement of ponds at multiple scales using Moran Eigenvector Maps (MEMs), testing for SAC with Moran correlogram



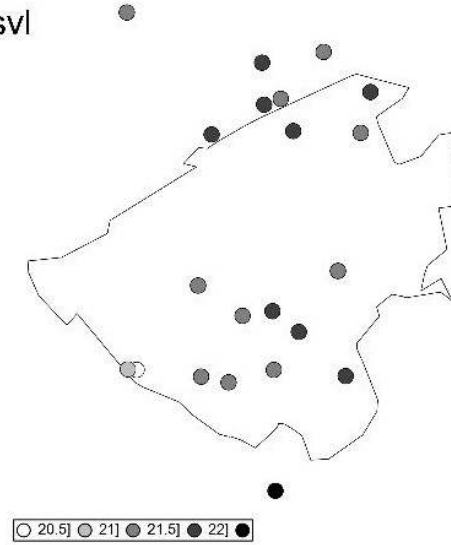




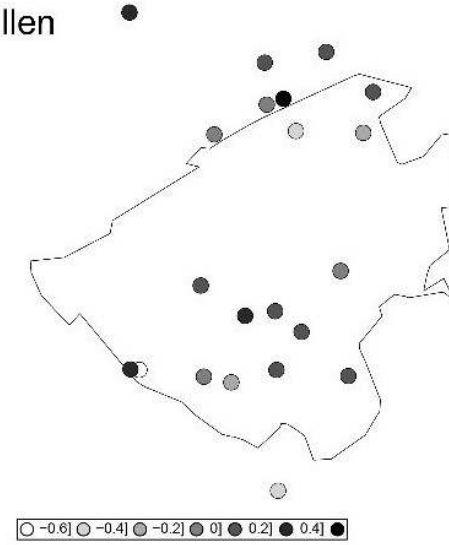


Not a clear spatial pattern
in mean phenotypic traits

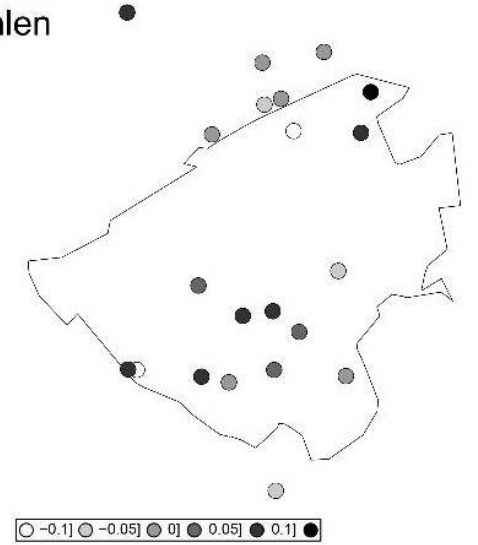
svl



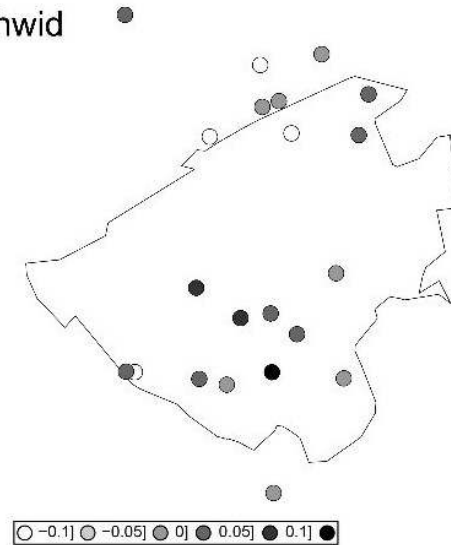
llen



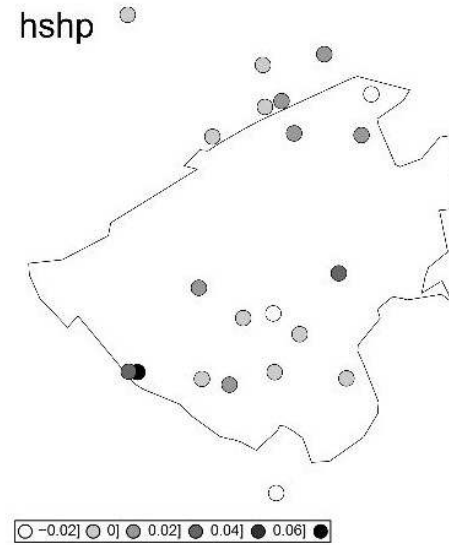
hlen



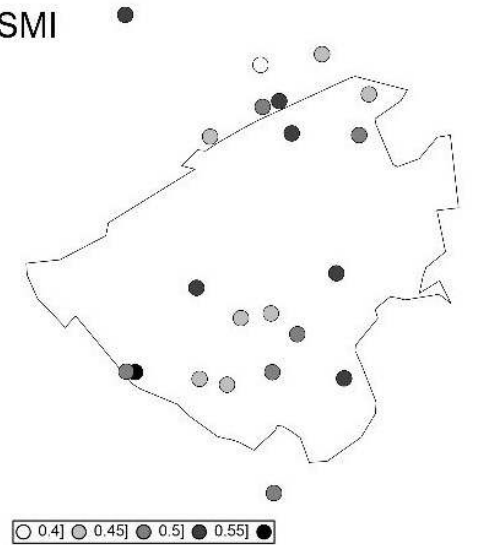
hwid



hshp



SMI



Discussion

Lack of a clear spatial pattern in phenotypic variation can be due to **weak selection gradient**, as a result of recent urbanisation;

Furthermore, there still could be moderate gene flow connecting populations, preventing local adaptation and fixation of phenotypes;

Discussion

Eco-evolutionary dynamics in tropical cities seem to have a different pace compared to temperate ones, which found marked changes in phenotypes of frogs between rural and urban sites;

Our results can contribute for building of a broader theory in urban ecology that explicitly includes the age, development mode, growth rate and history of occupation of cities;

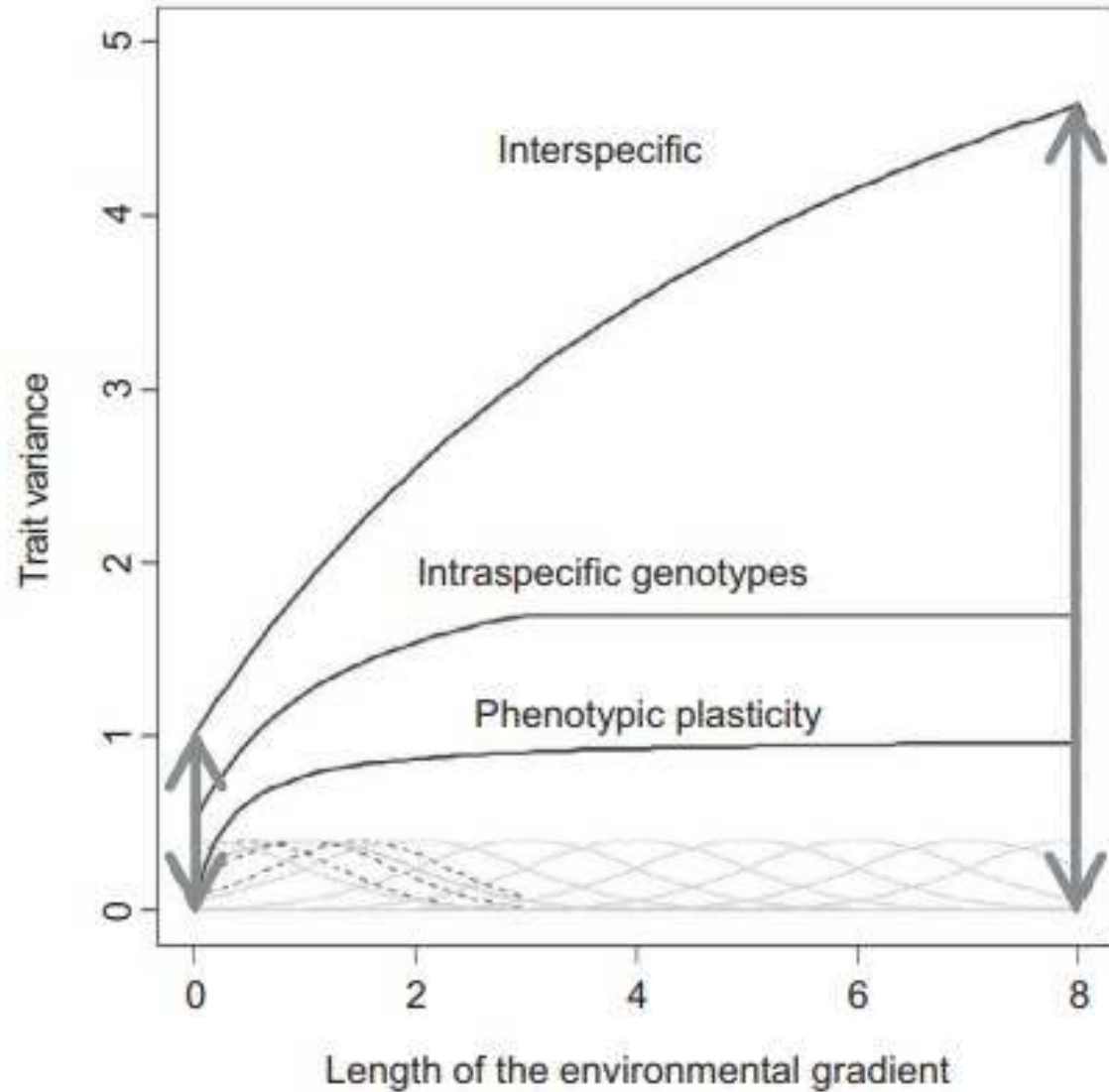
Finally, urbanisation seems to act indirectly via changes in environmental heterogeneity



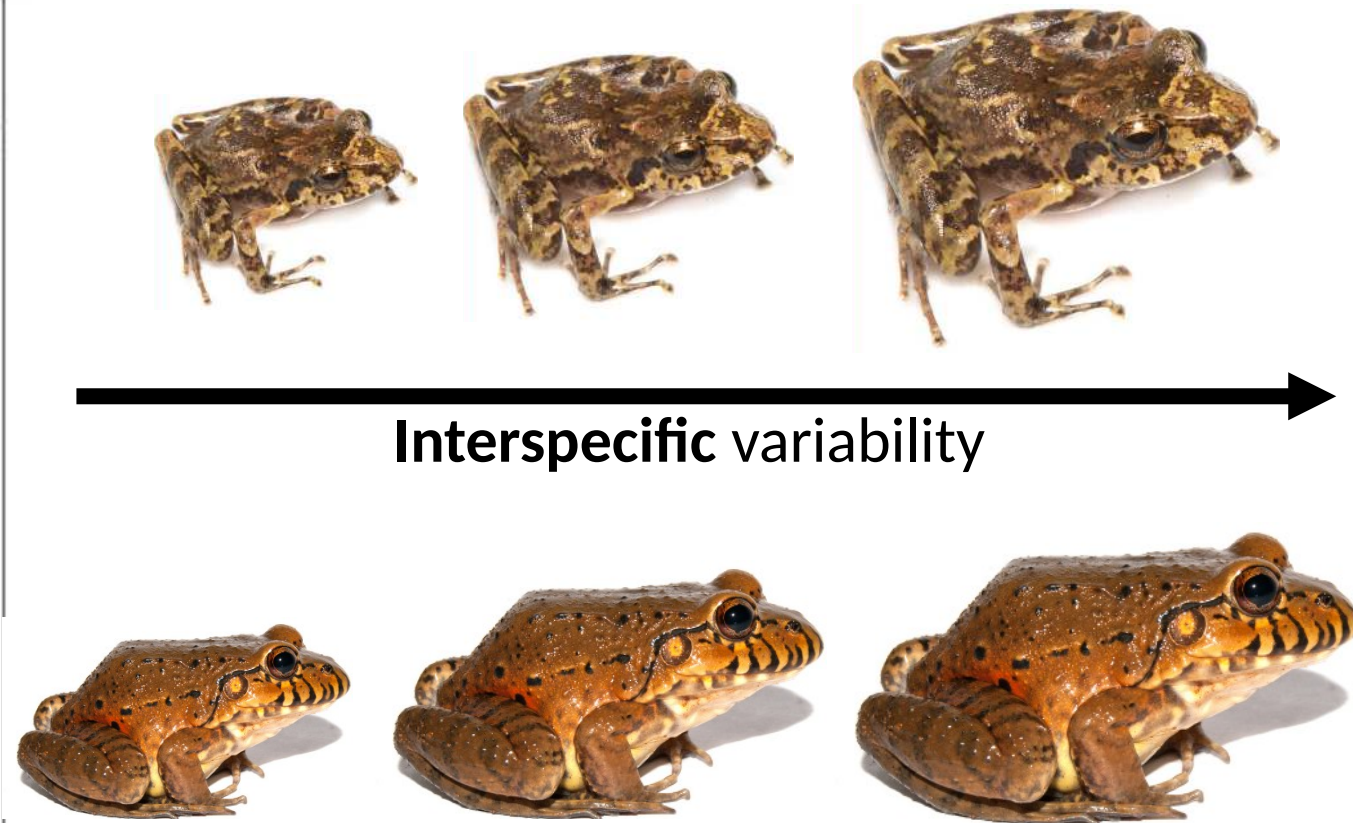
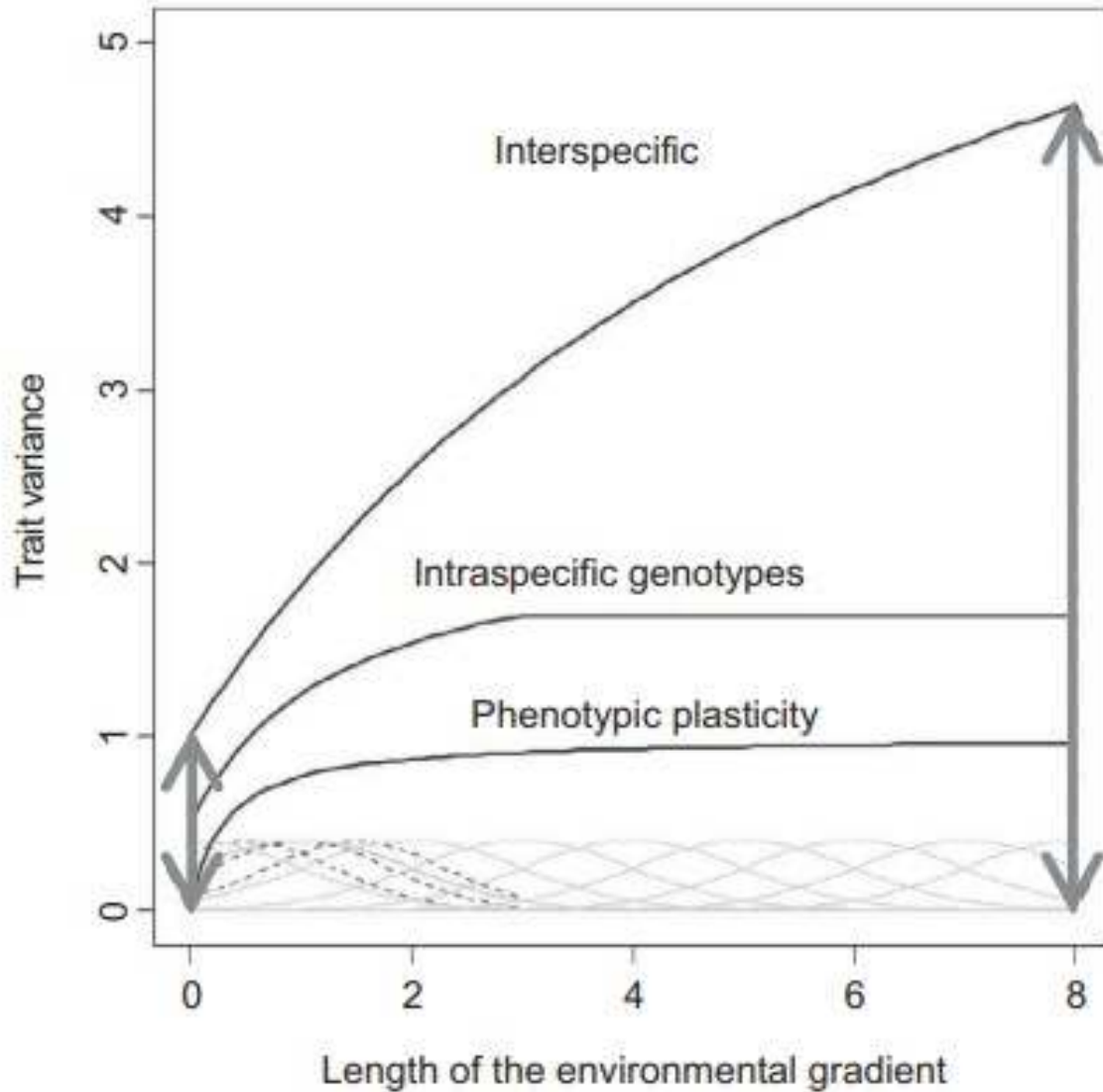
Leandro B Menezes
Master's candidate

Effects of urbanisation
on intra- and interspecific
phenotypic variation
in two metacommunities

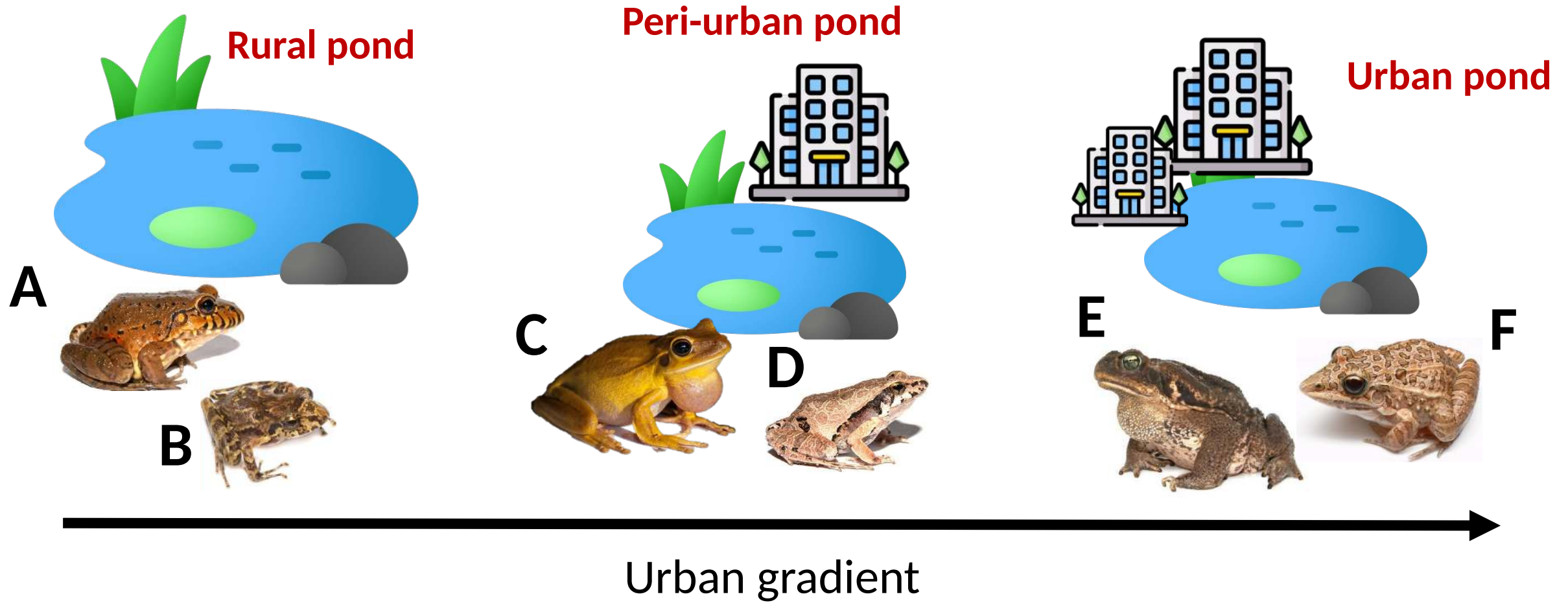
Intraespecific and interspecific variability



Intraespecific and interspecific variability

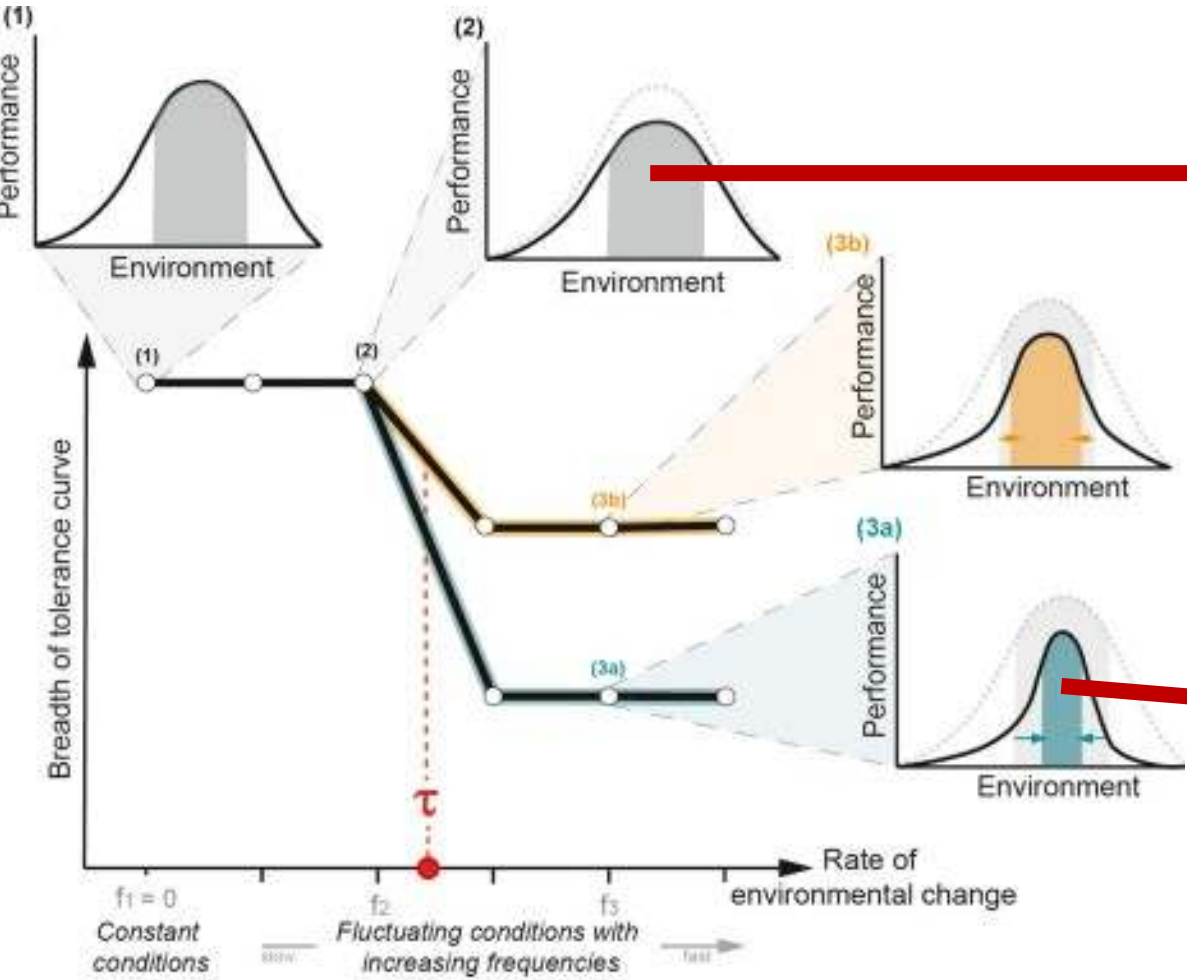


Phenotypic variability due to species turnover



Stablizing selection at population level

Introduction

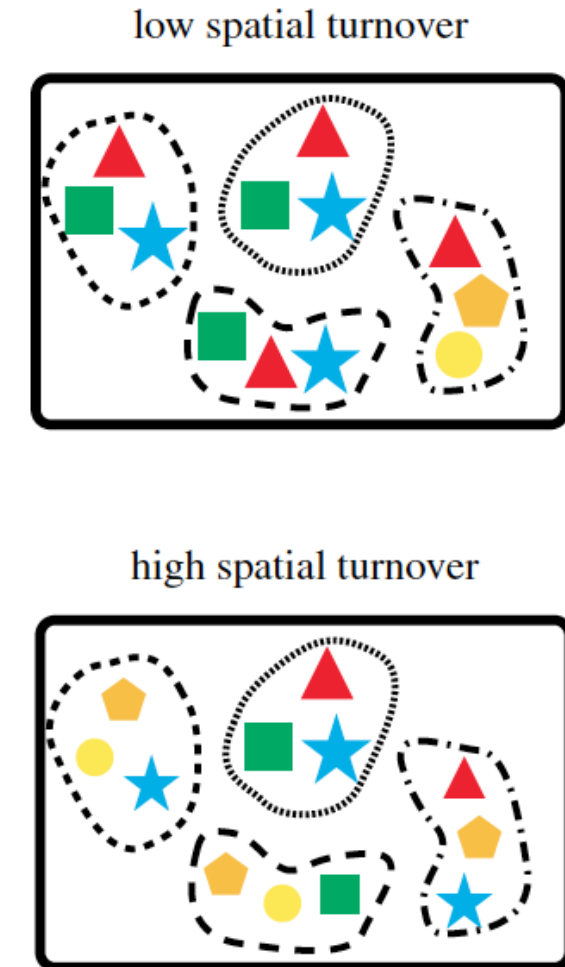
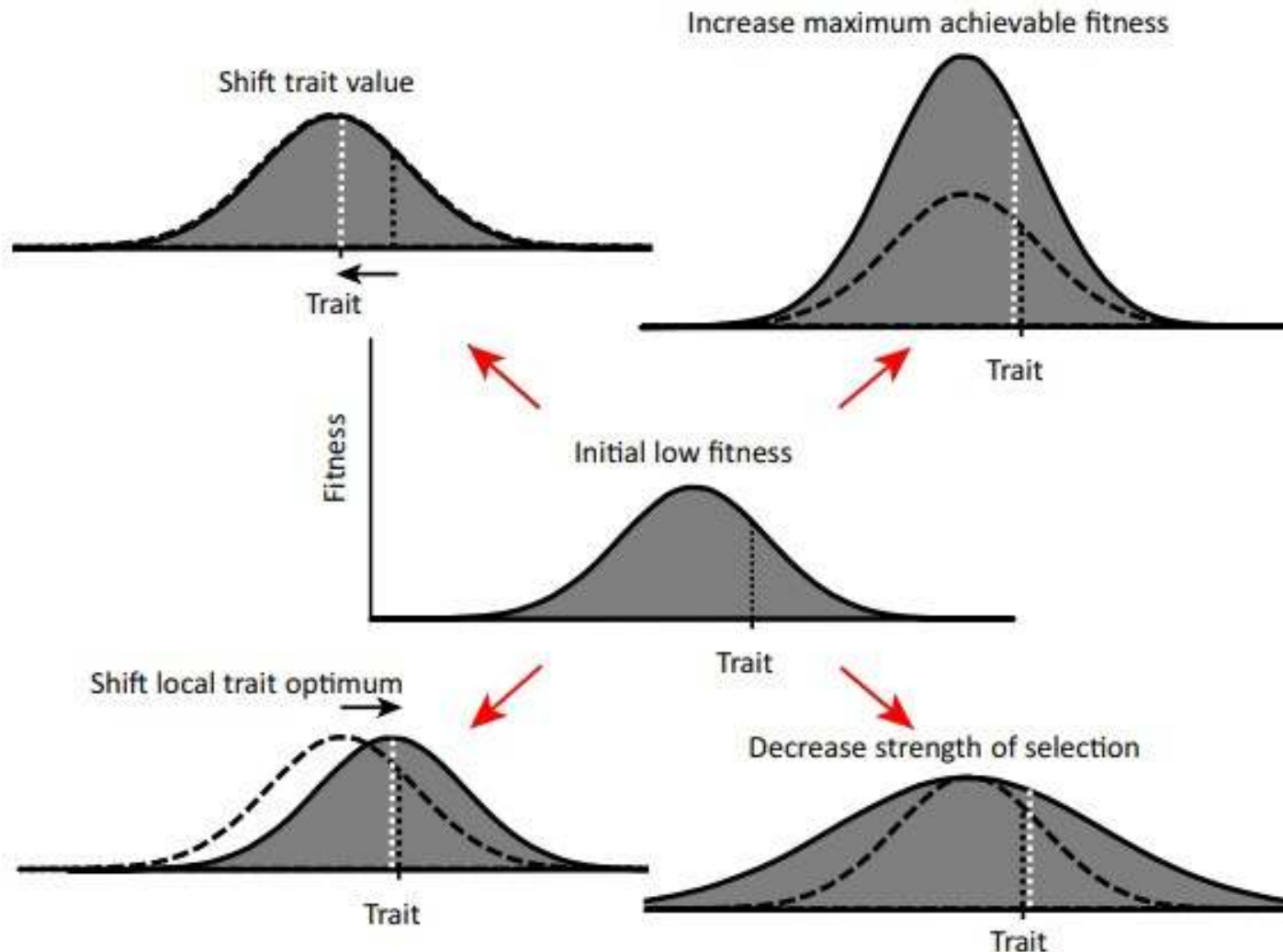


High phenotypic variability



Low phenotypic variability

Low phenotypic variability generally implies in **loss** of fitness



Methods

Piano et al. 2020

Received: 28 March 2019 | Accepted: 1 November 2019

DOI: 10.1111/gcb.14934

PRIMARY RESEARCH ARTICLE

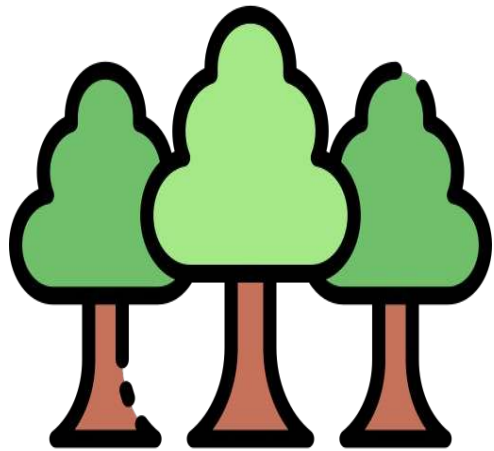
Global Change Biology WILEY

Ecological and
evolutionary factors

+

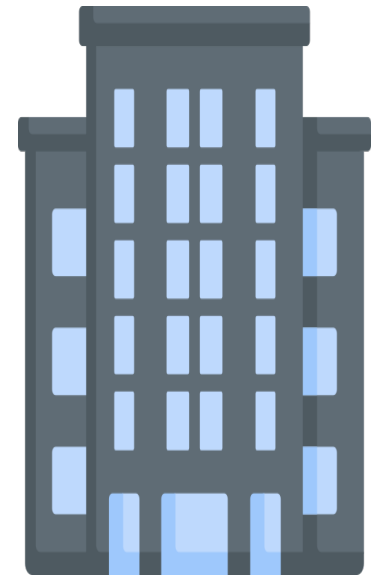
Environment

Urbanization drives cross-taxon declines in abundance and
diversity at multiple spatial scales



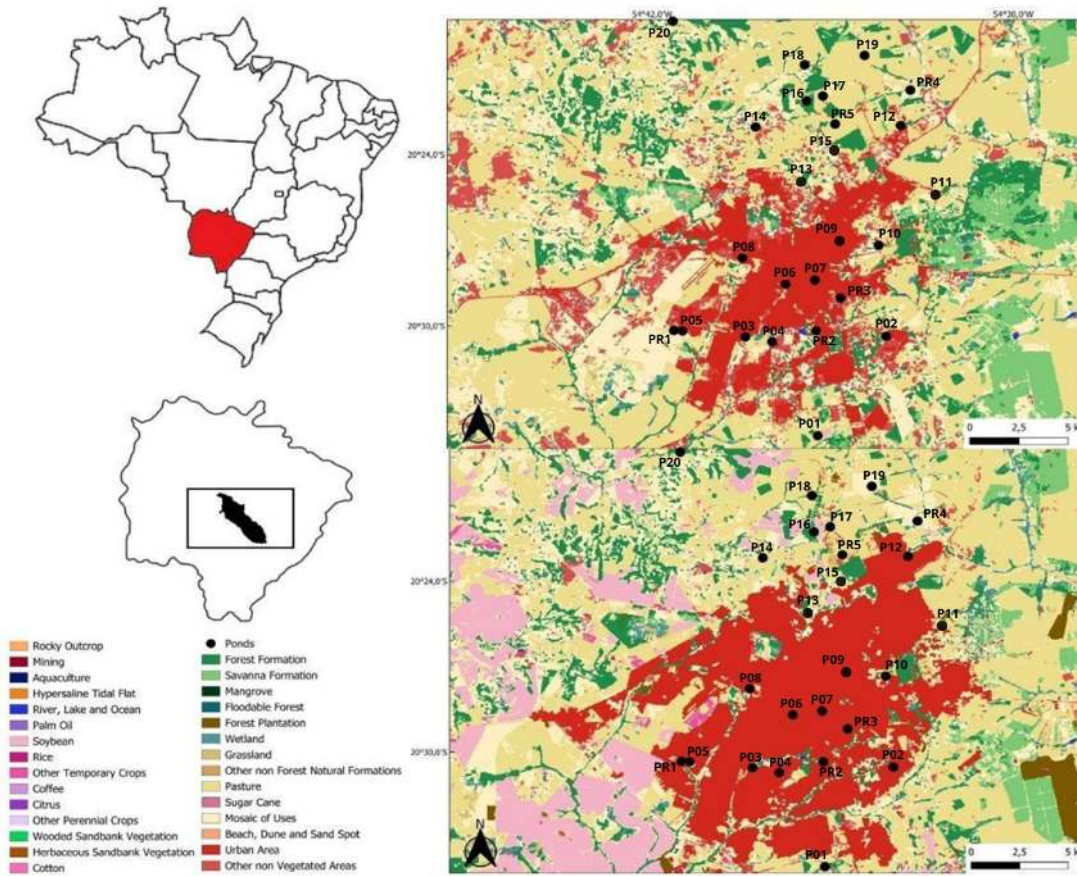
Urbanisation (%)

20 ponds (500-m buffer)



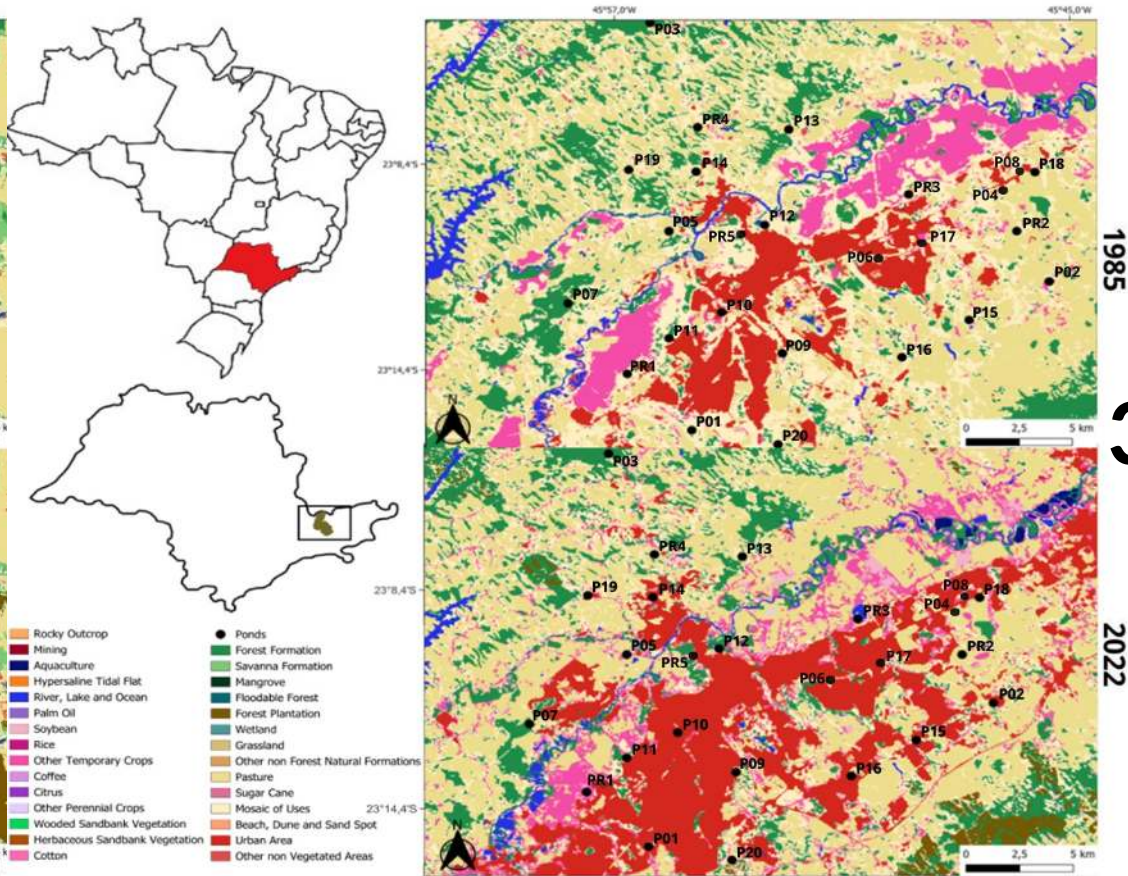
Study areas

Campo Grande (MS)

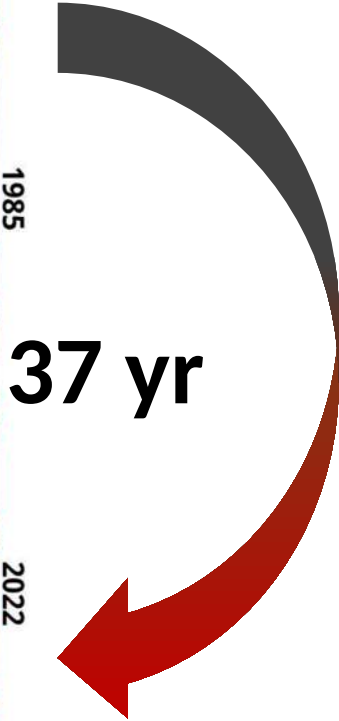


Founded 124 yr ago

São José dos Campos (SP)

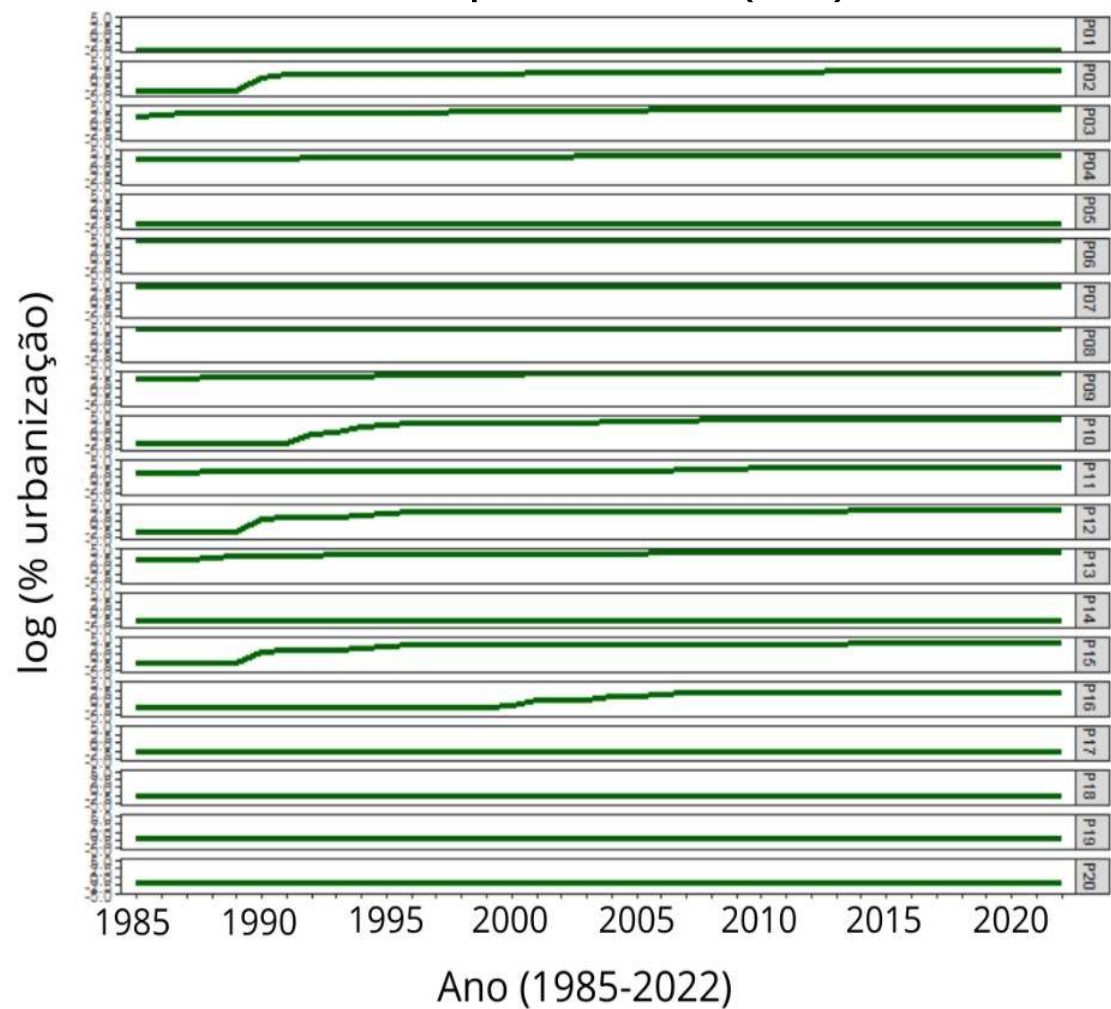


256 yr ago

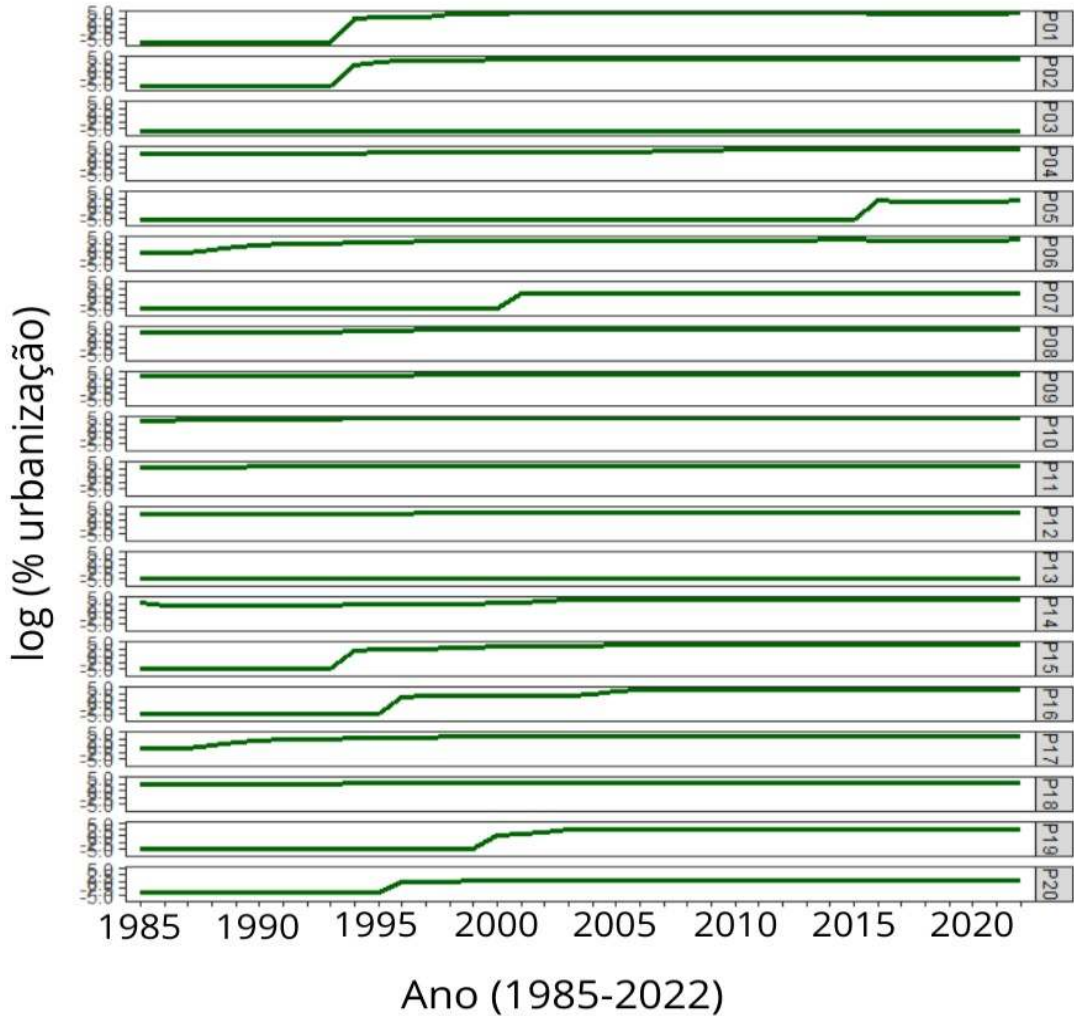


Time series of urbanisation *per pond*

Campo Grande (MS)



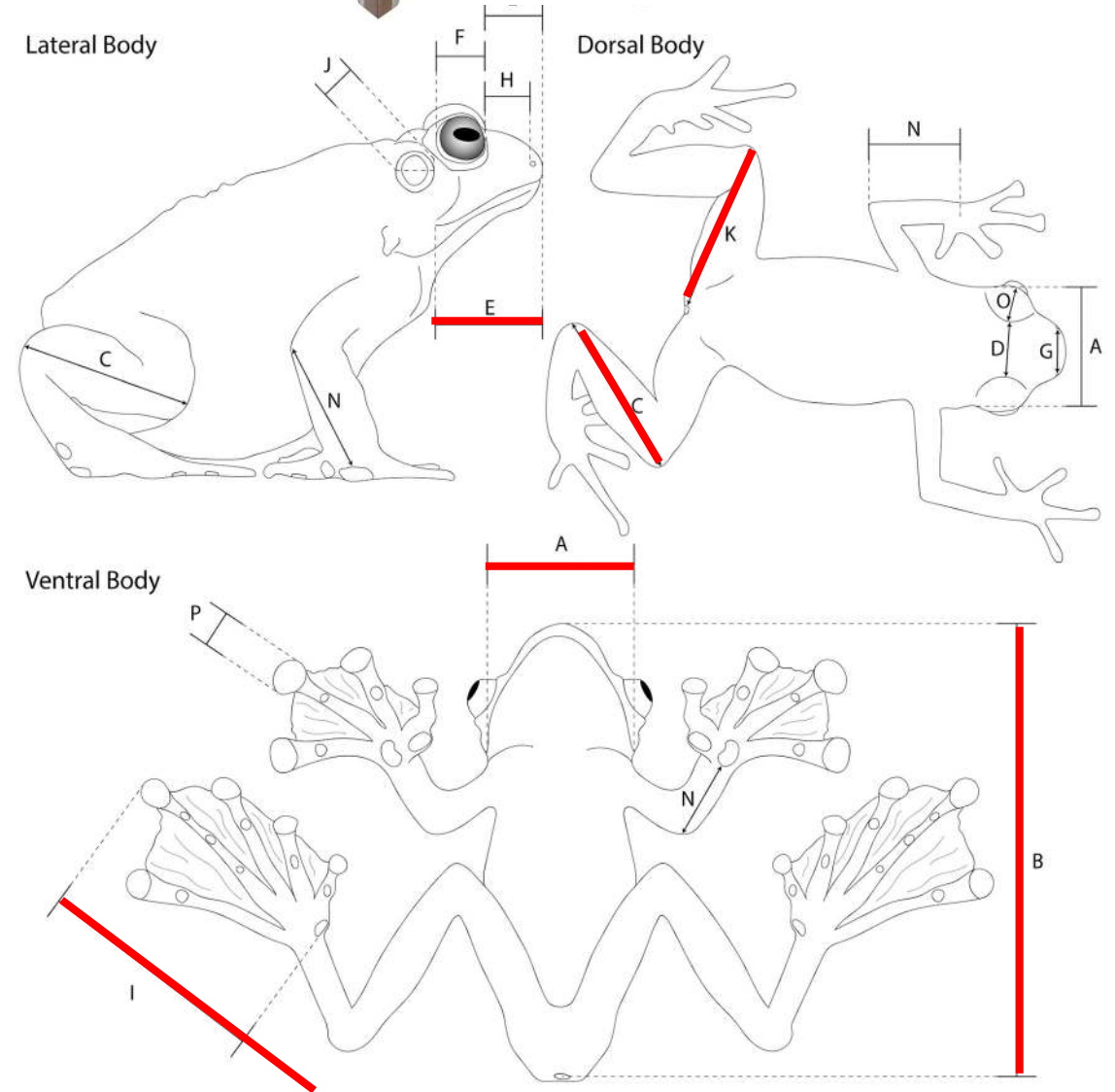
São José dos Campos (SP)



Phenotypic traits



Left side of adult males
(5 individuals per **species** per
pond)



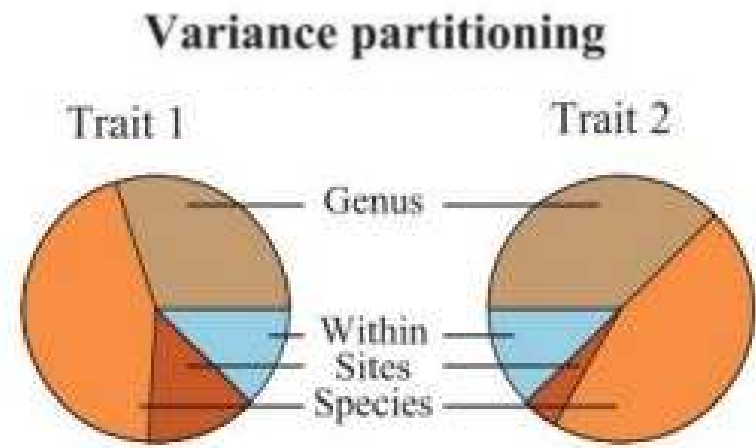
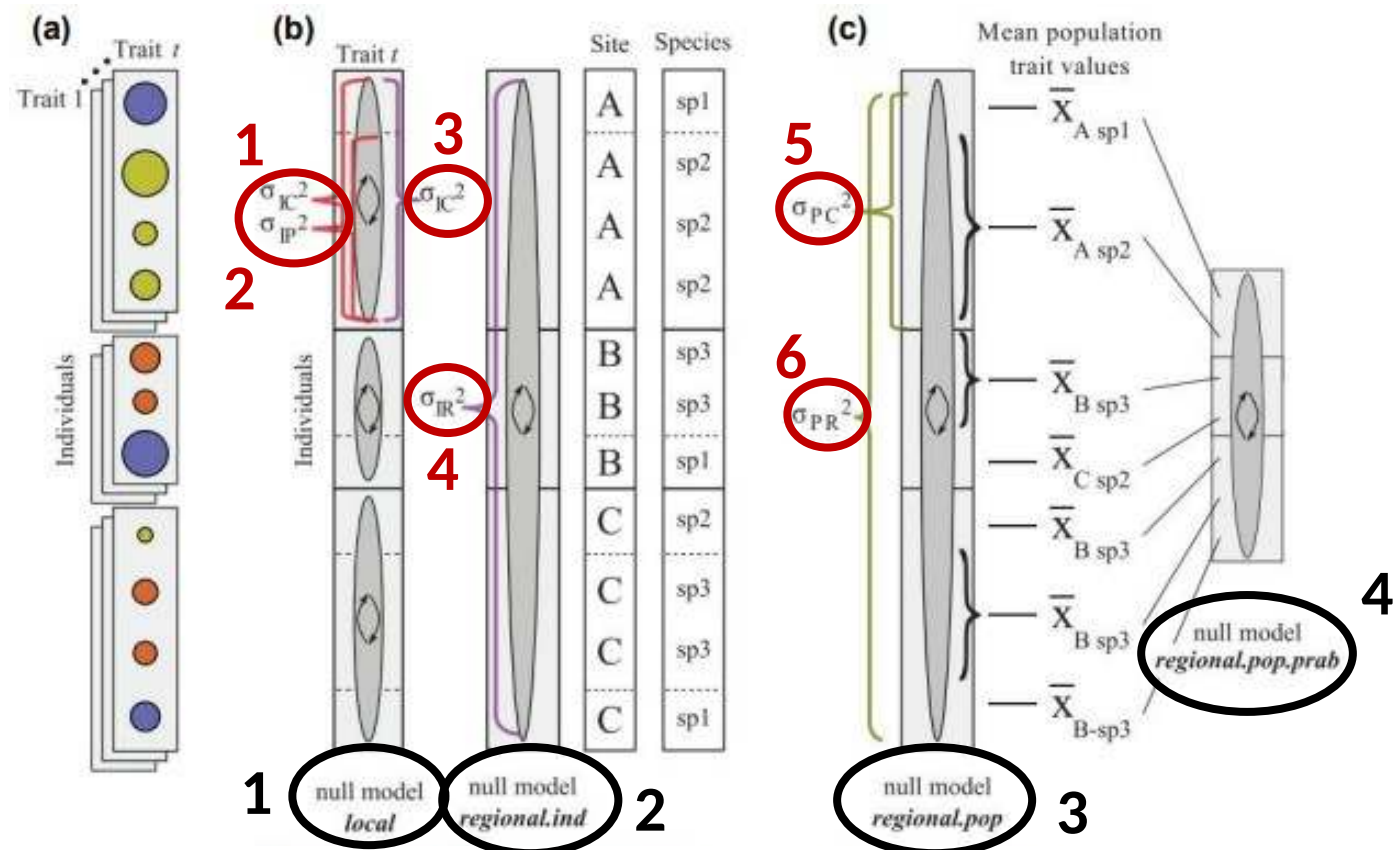
1st question

Do urban ecosystems constrain **phenotypic variation** related to **distinct niche components**?

Data analysis

1) Test how the **urbanisation gradient** changes the contribution of **intraspecific phenotypic variation** in relation to interspecific;

Package *cati* 



Data analysis

- 1) Test how the **urbanisation gradient** changes the contribution of **intraspecific phenotypic variation** in relation to interspecific;

T statistics (variation partitionin akin of *F*)

1

$$T_{IP/IC} = \frac{\sigma_{IP}^2}{\sigma_{IC}^2}$$

IP = Intraspecific variation at the population
IC = Intraspecific variation at the community scale

2

$$T_{IC/IR} = \frac{\sigma_{IC}^2}{\sigma_{IR}^2}$$

IC = Intraspecific variation at the community scale

IR = Variação intra- no pool regional

3

$$T_{PC/PR} = \frac{\sigma_{PC}^2}{\sigma_{PR}^2}$$

PC = Variação das médias na população

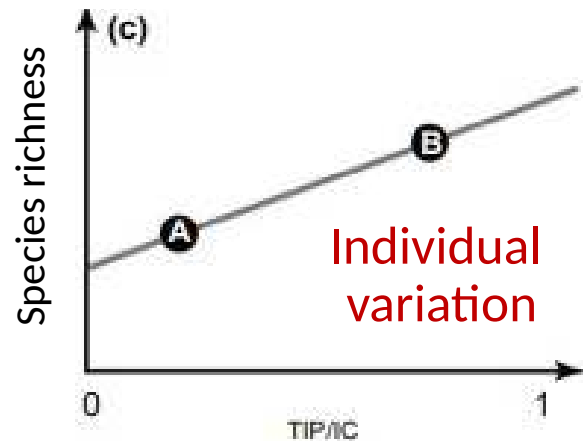
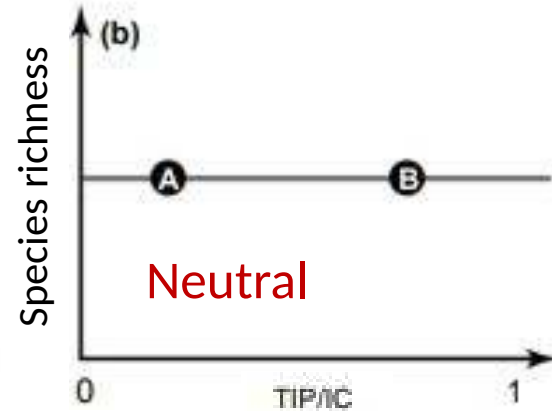
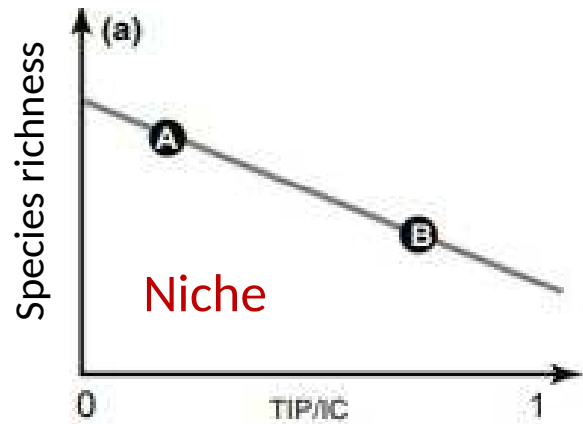
PR = Variação das médias na comunidade

Pacote *cati*



Data analysis

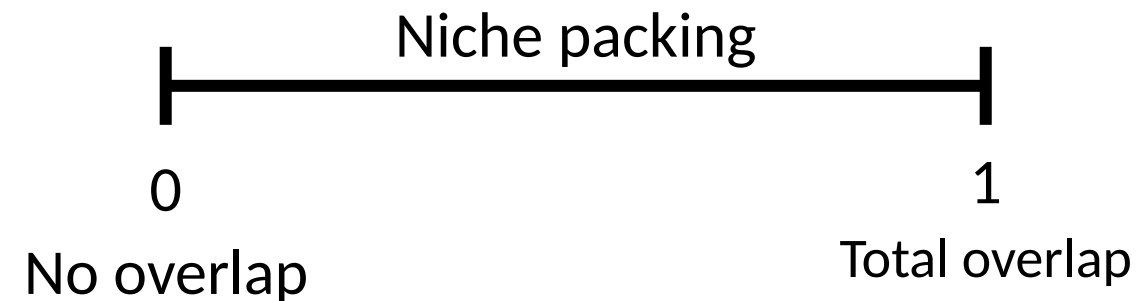
1) Test how the **urbanisation gradient** changes the contribution of **intraspecific** in relation to **interspecific phenotypic** variation;



Package *cati* 

within-population variance conditional
within the community

$$T_{IP/IC} = \frac{\sigma_{IP}^2}{\sigma_{IC}^2}$$



Results

1.051 Individuals measured

5 Families

44 Species

Campo Grande: **27** Species

São José dos Campos: **26** Species



Dendropsophus (24.69%)



Leptodactylus (19.22%)



Boana (17.41%)



Scinax (15.32%)

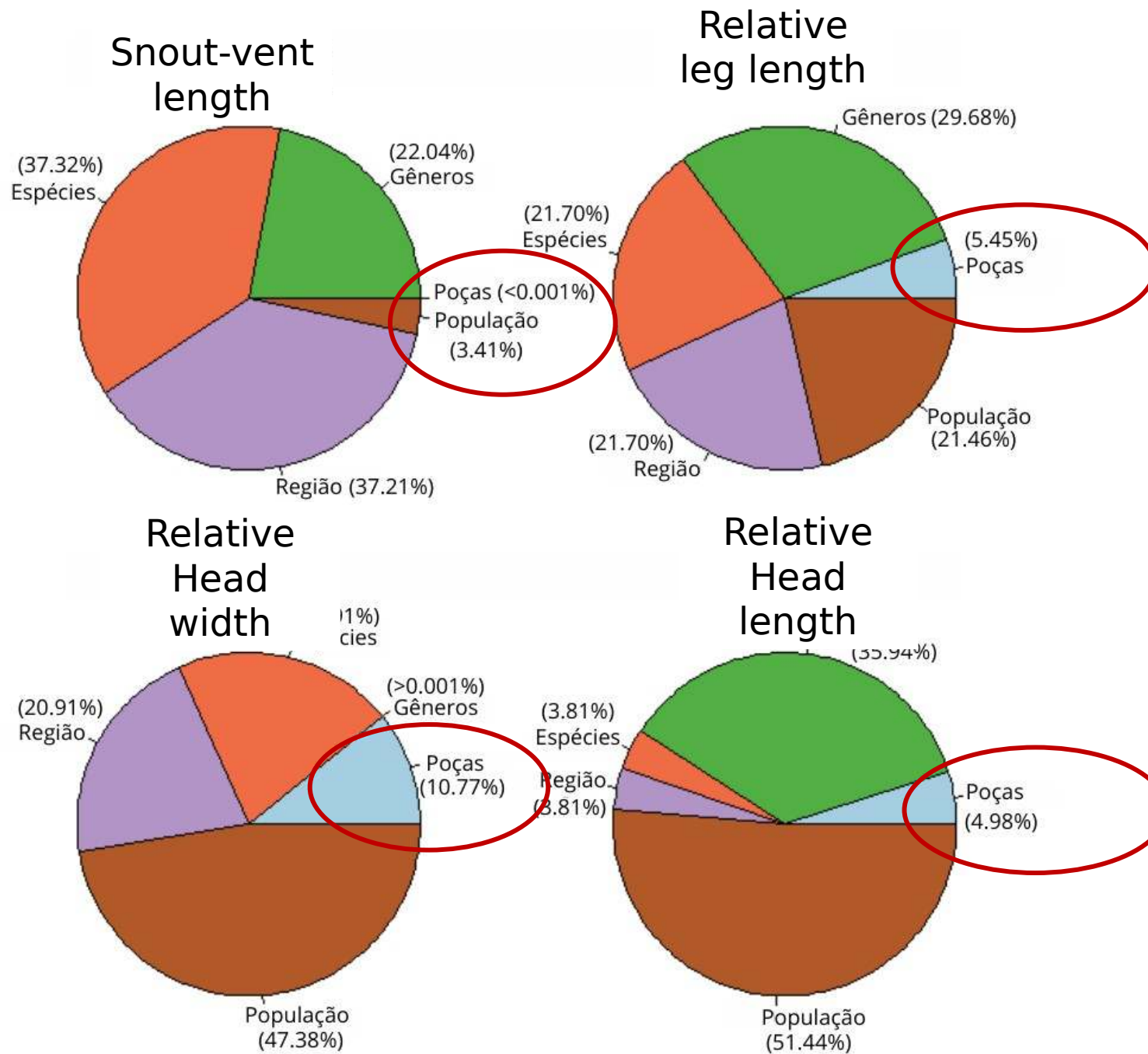


Physalaemus (11.99%)

Hypothesis 1) Urbanisation constrains the **variation in phenotypic traits** related to the **impact** and requirement niche;

Phenotypic variation was **higher** at the genera, species, and city level, but **lower** at the population level.

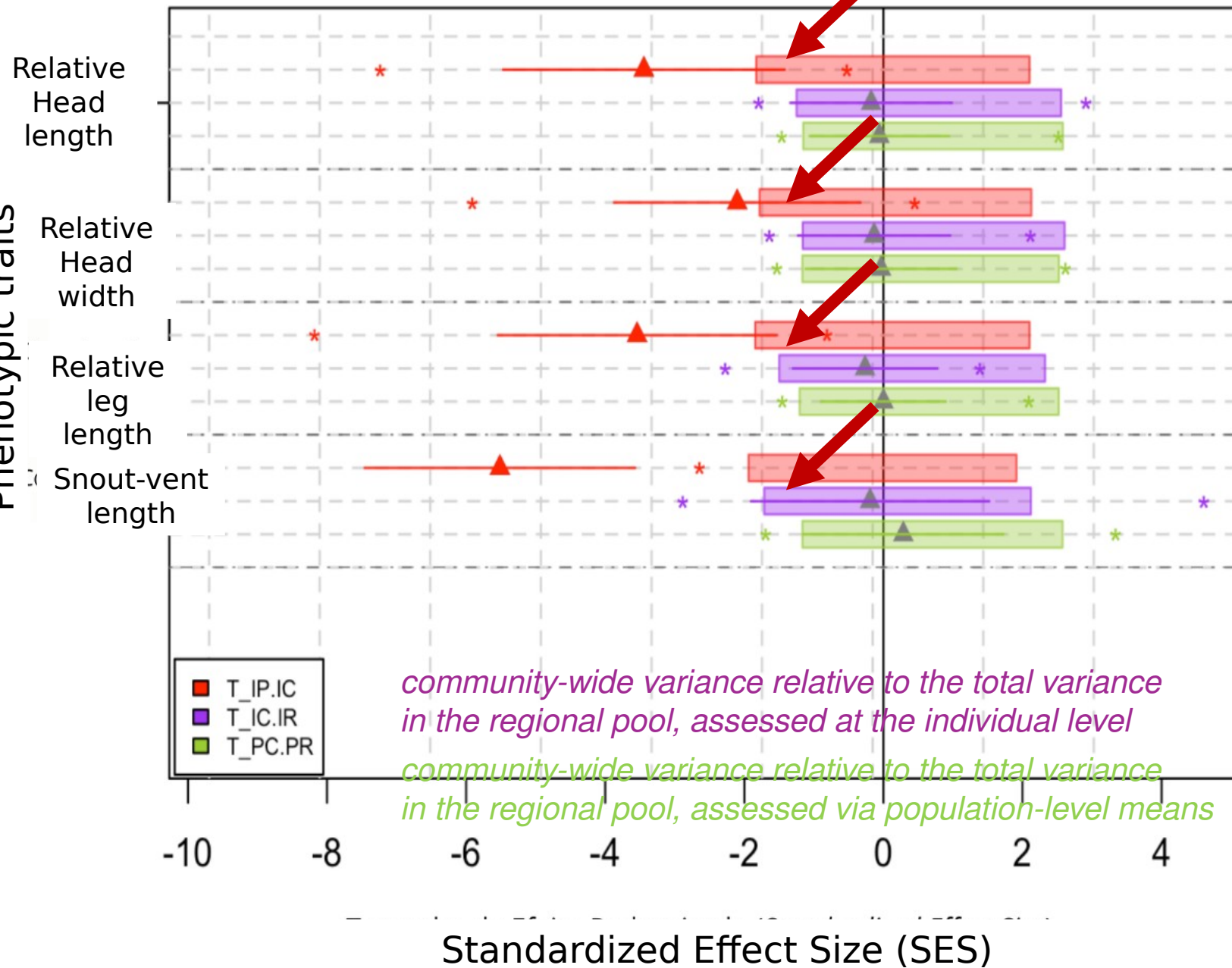
Results



Low intraspecific variation among ponds along the urbanisation gradient

Results

Phenotypic traits

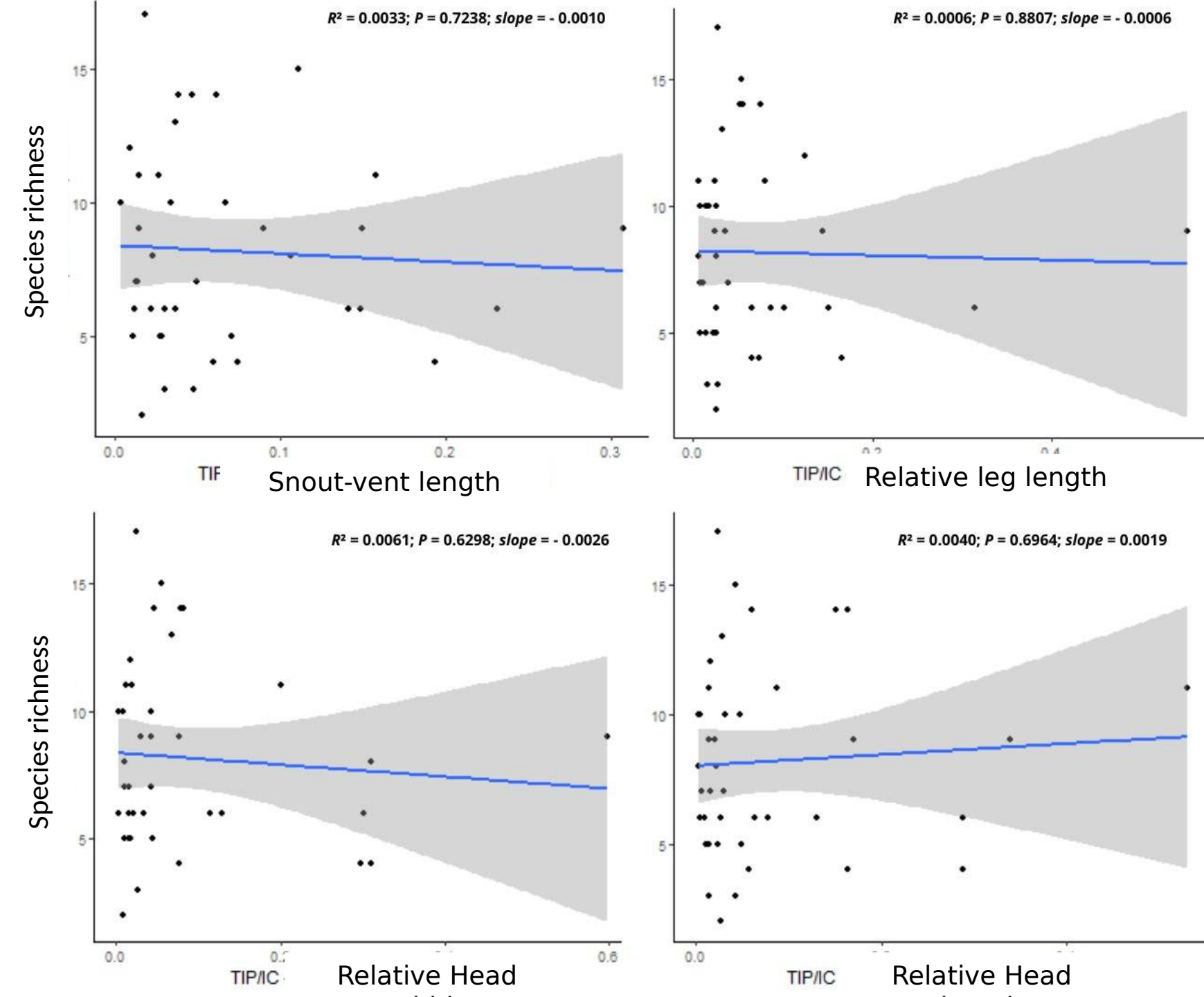


Red dots are actual distribution of the data. Boxes are null distribution (null model). The farthest red dots are from the box, the more significant the statistics

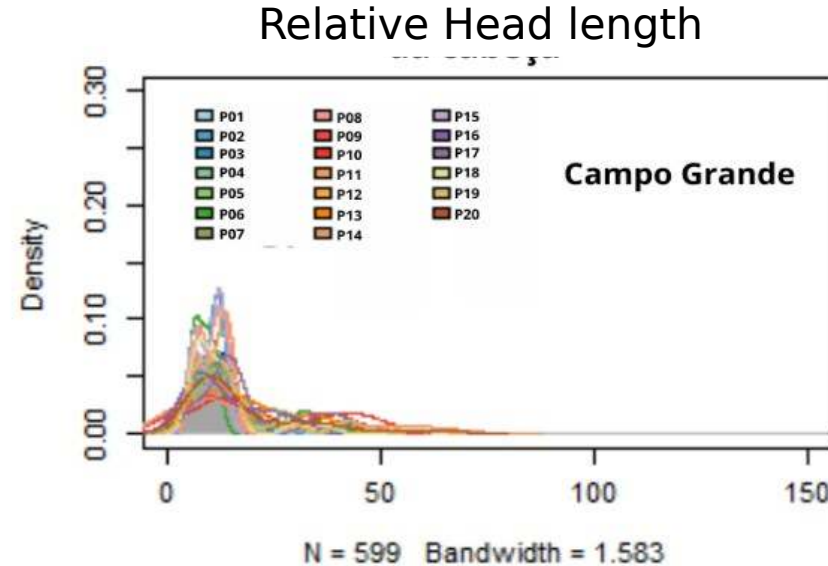
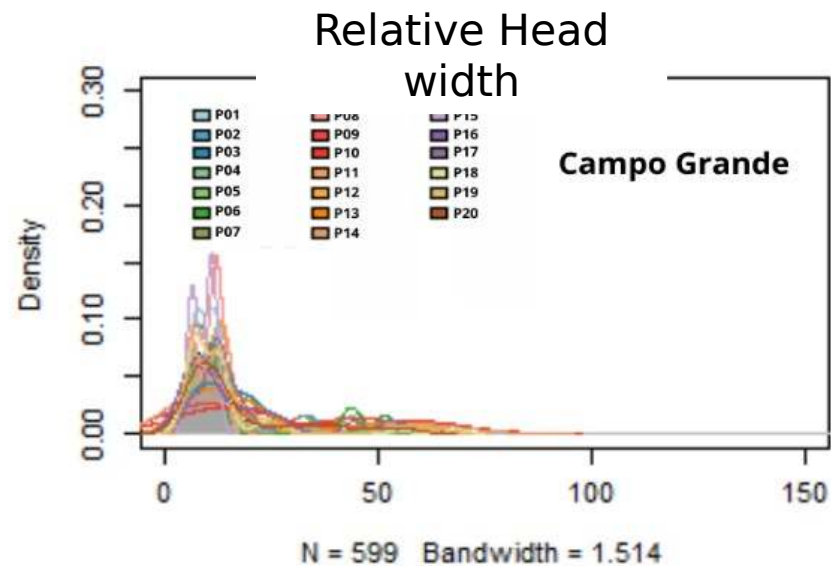
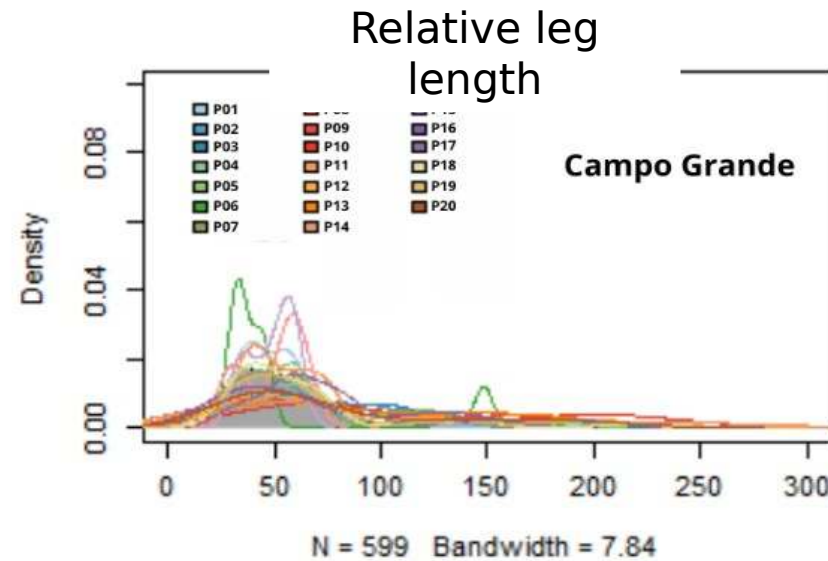
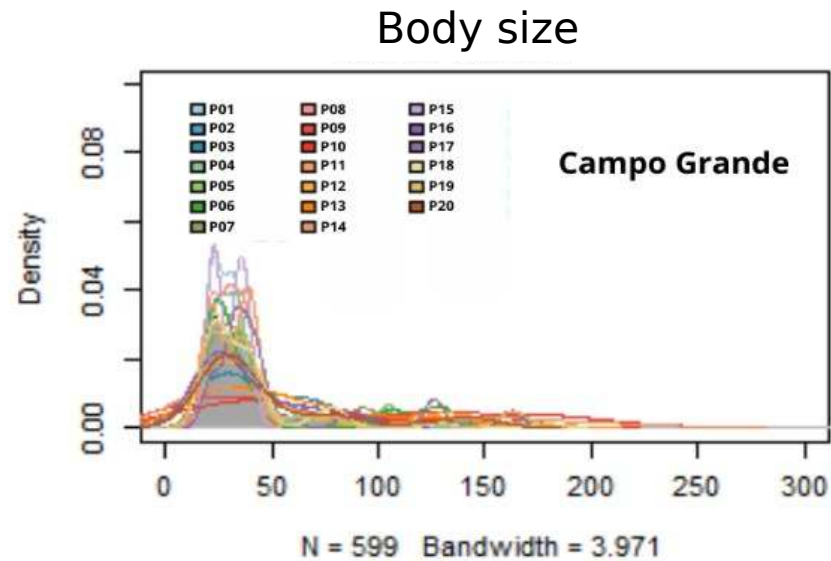
Not much variability at the community level relative to the regional pool, either measured at the population or individual level

Results

Neutral processes shape
communities



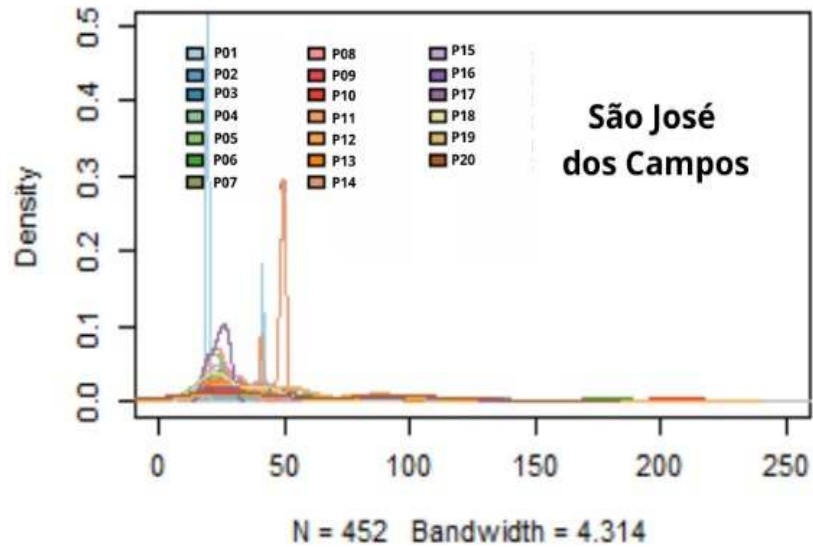
Results



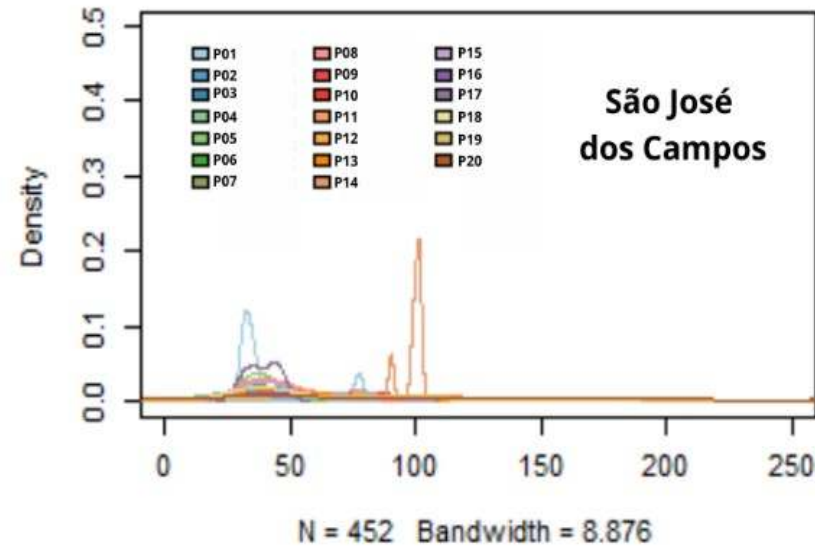
High overlap
in phenotypic variation
among ponds
in younger city

Results

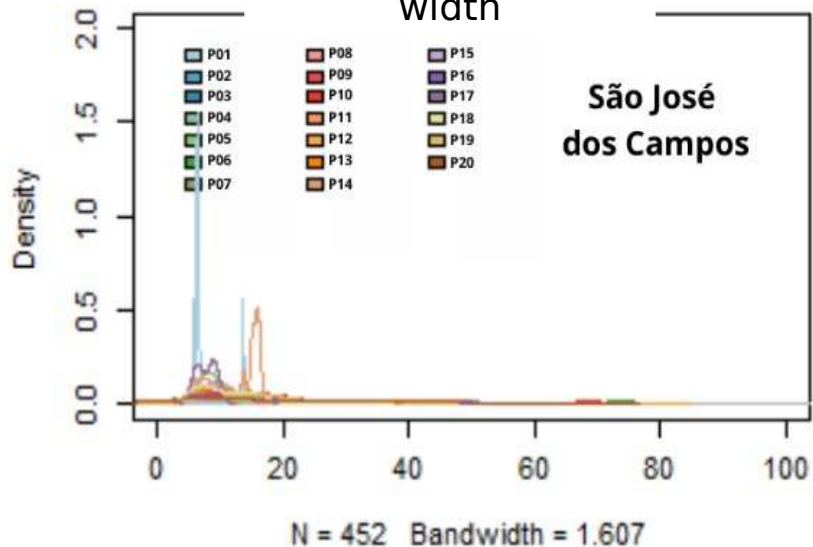
Comprimento
rostro-cloacal



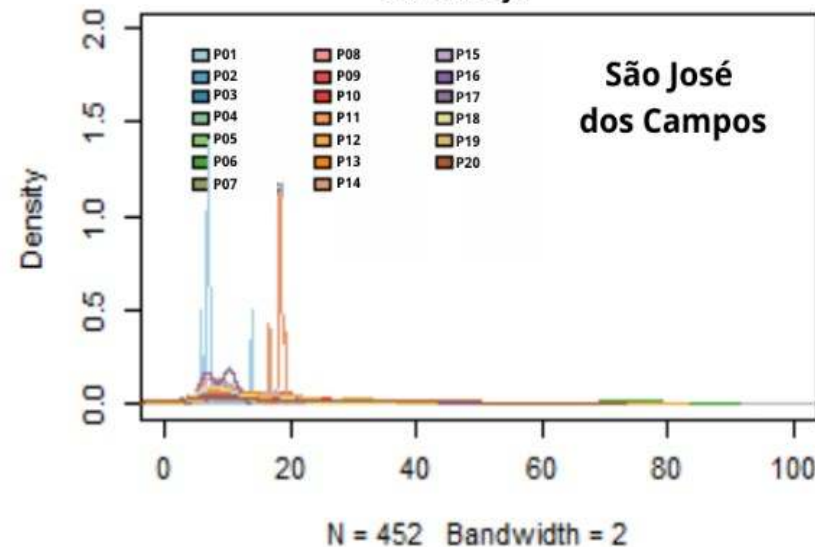
Comprimento
da perna



Relative Head
width



Comprimento
da cabeça



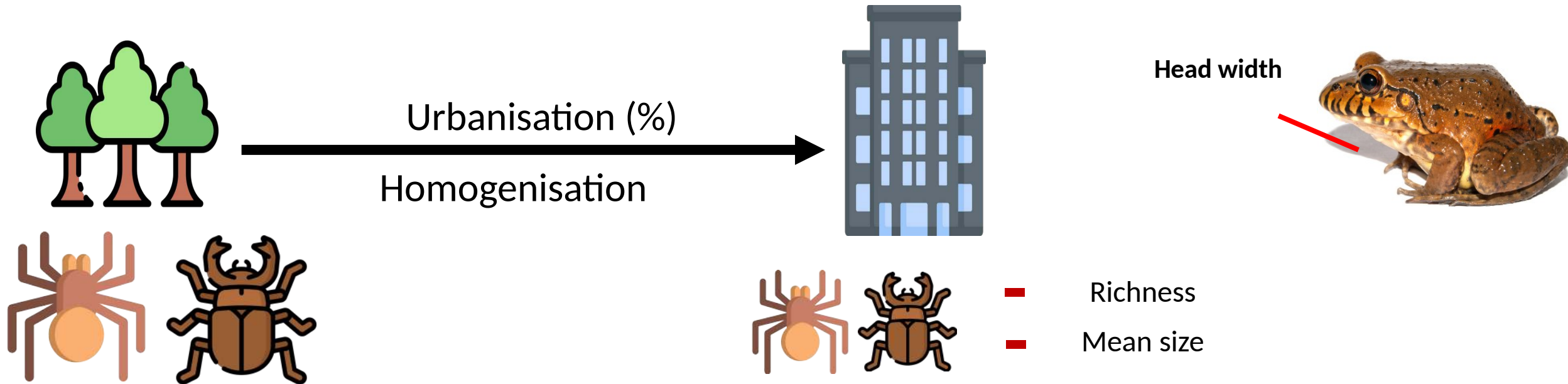
Low overlap

in phenotypic variation
among ponds
in the **older city**

Discussion

Urban ecosystems seem to experience wide **phenotypic homogenisation** at the scale of communities

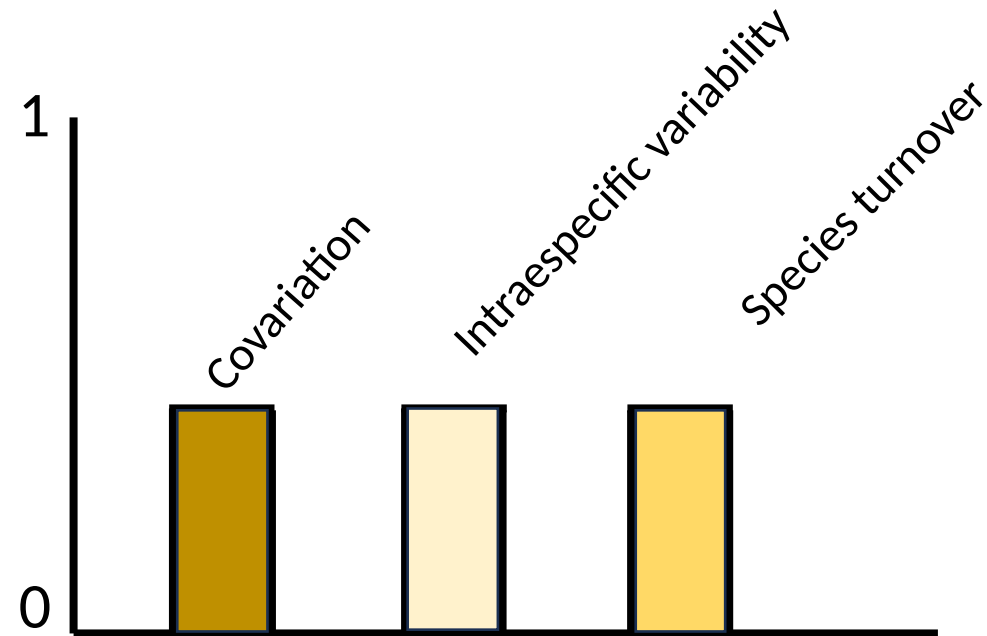
Homogenisation of **feeding resources** may have driven to a low intraspecific variation in head width;



Data analysis

2) Test which local environmental variable + landscape composition best explains the **phenotypic variation** along the urbanisation gradient;

Package *cati*
Function *decompCTRE*



Data analysis

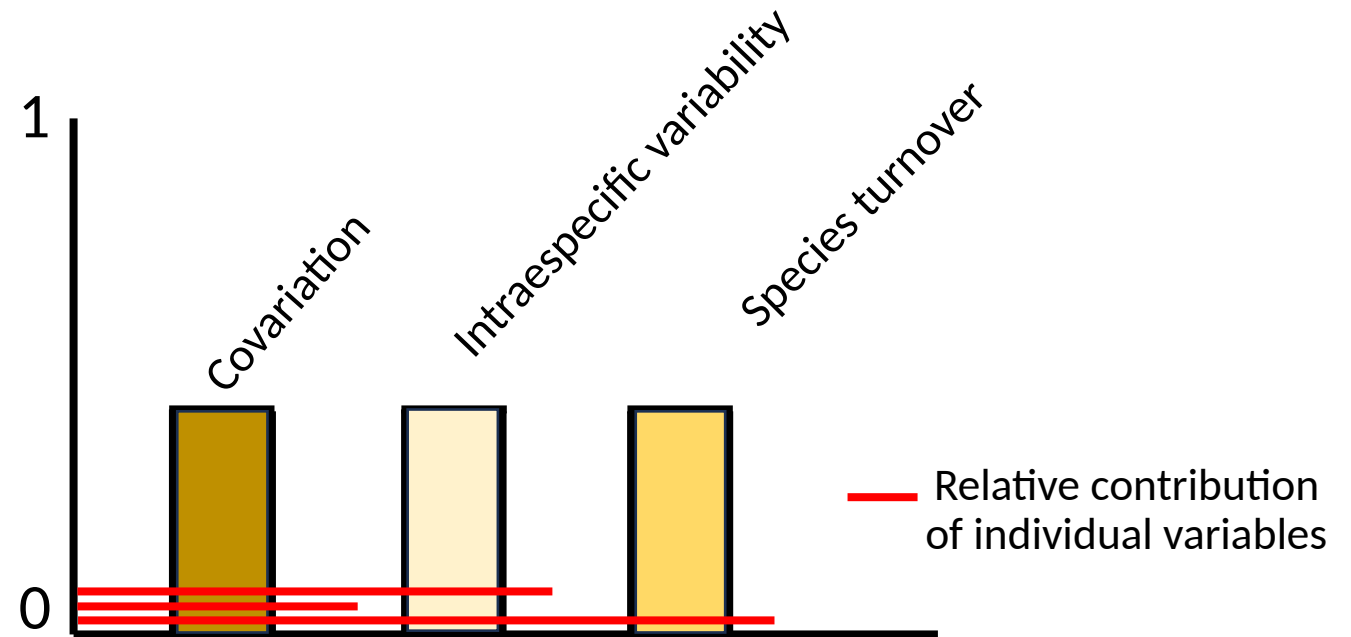
2) Test which local environmental variable + landscape composition best explains the **phenotypic variation** along the urbanisation gradient;

Package *cati*
Function *decompCTRE*



Linear Model

- PCs from landscape and local environmental variables

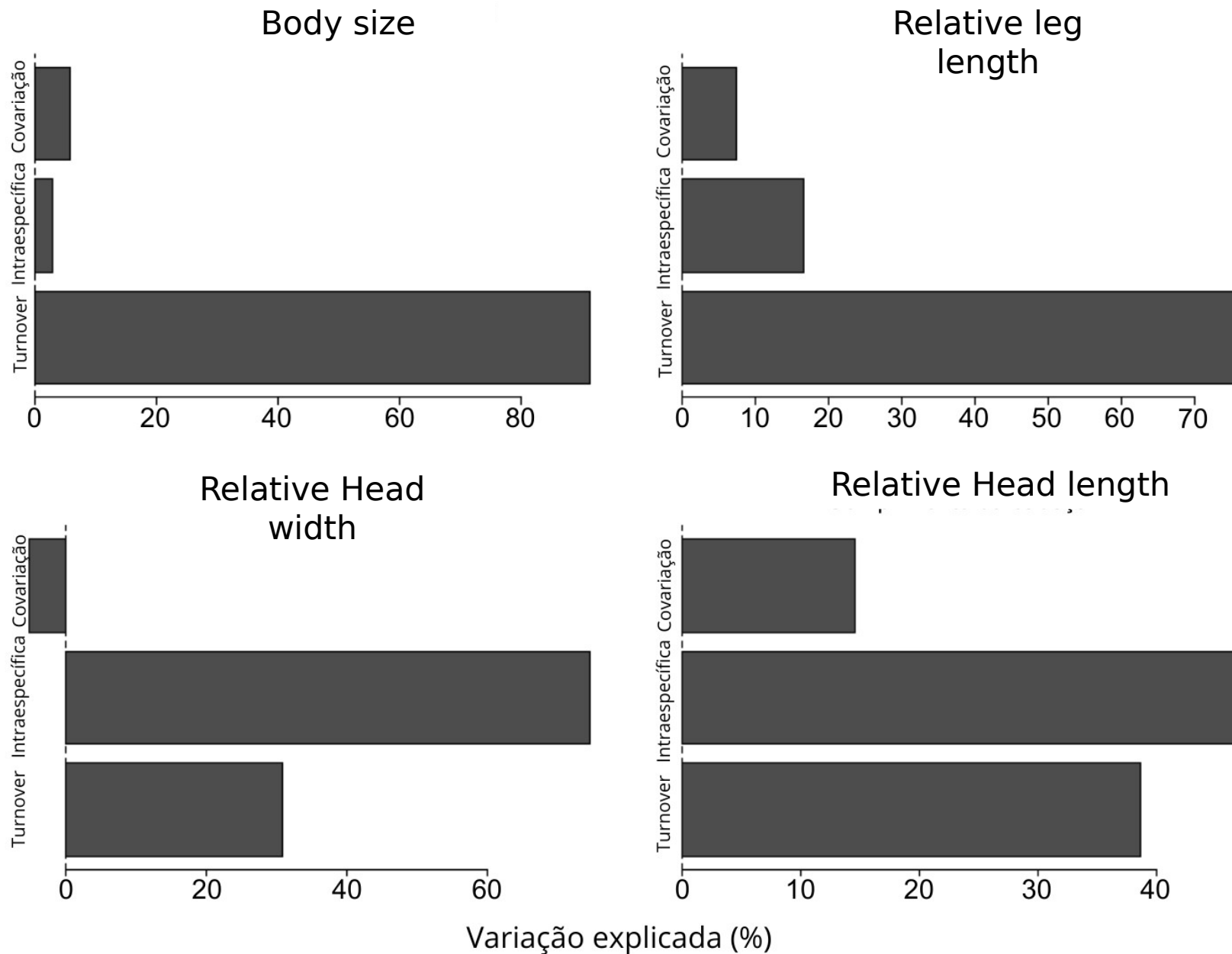


Hypothesis 2

2) **Phenotypic variability** will be more influenced by **local environmental variables** that are more closely related to the requirement niche, such as temperature and humidity;

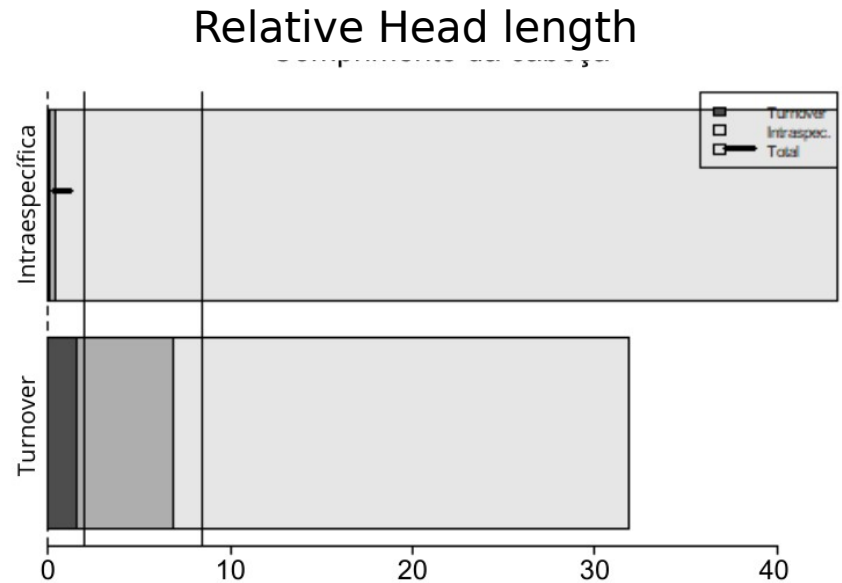
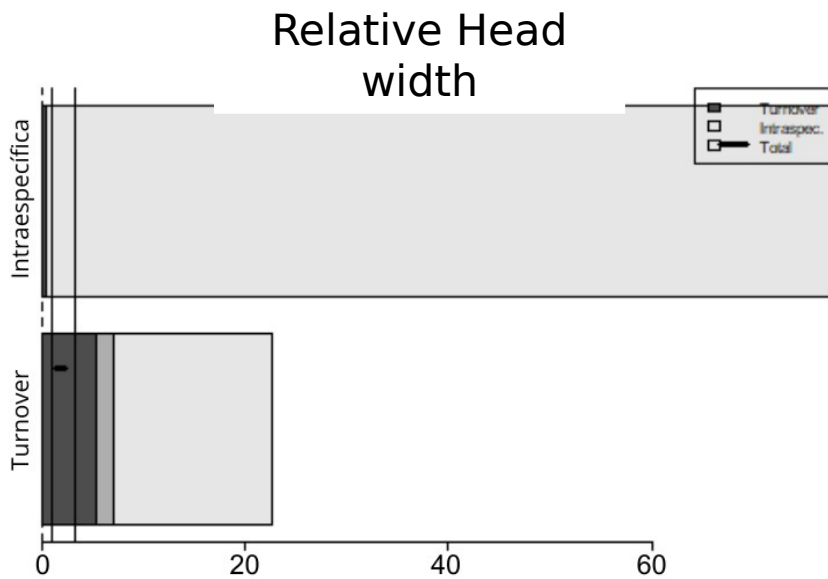
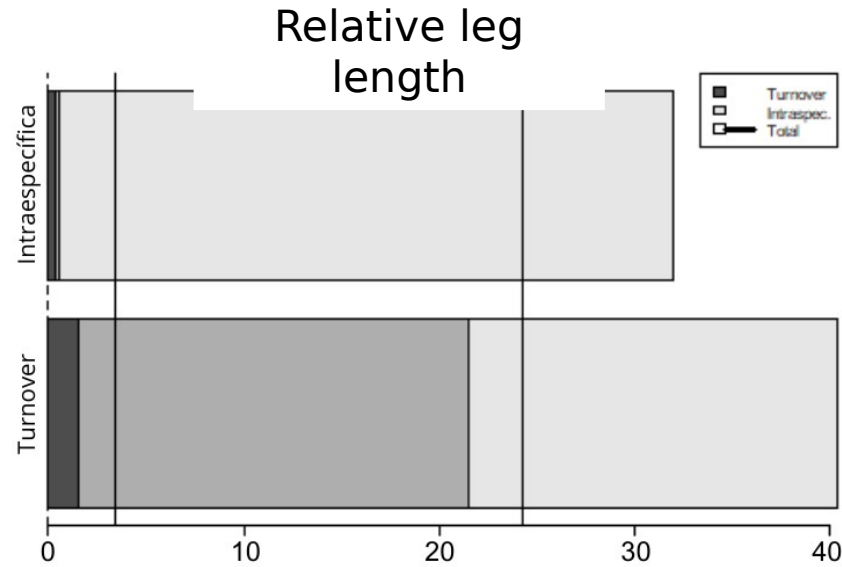
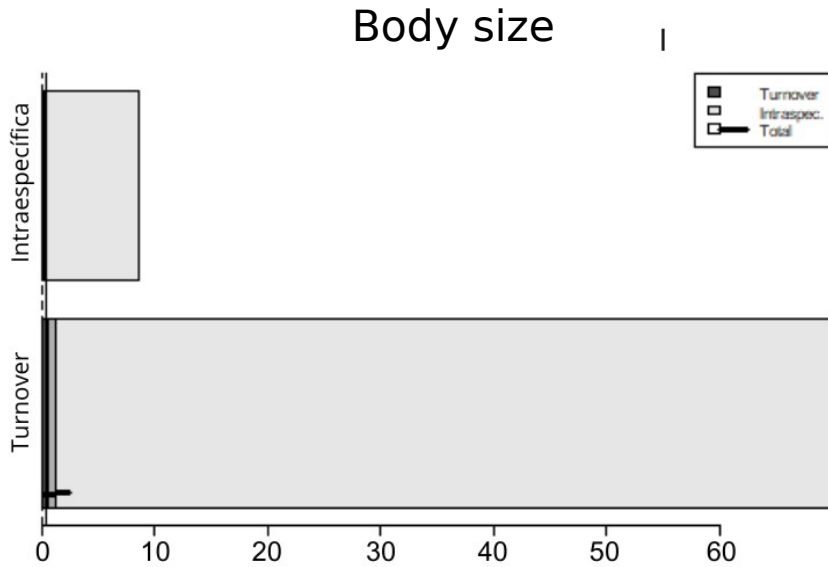


Results



Species turnover along the urbanisation gradient, instead of intraspecific variation is responsible for the larger part of total phenotypic variation in the older city

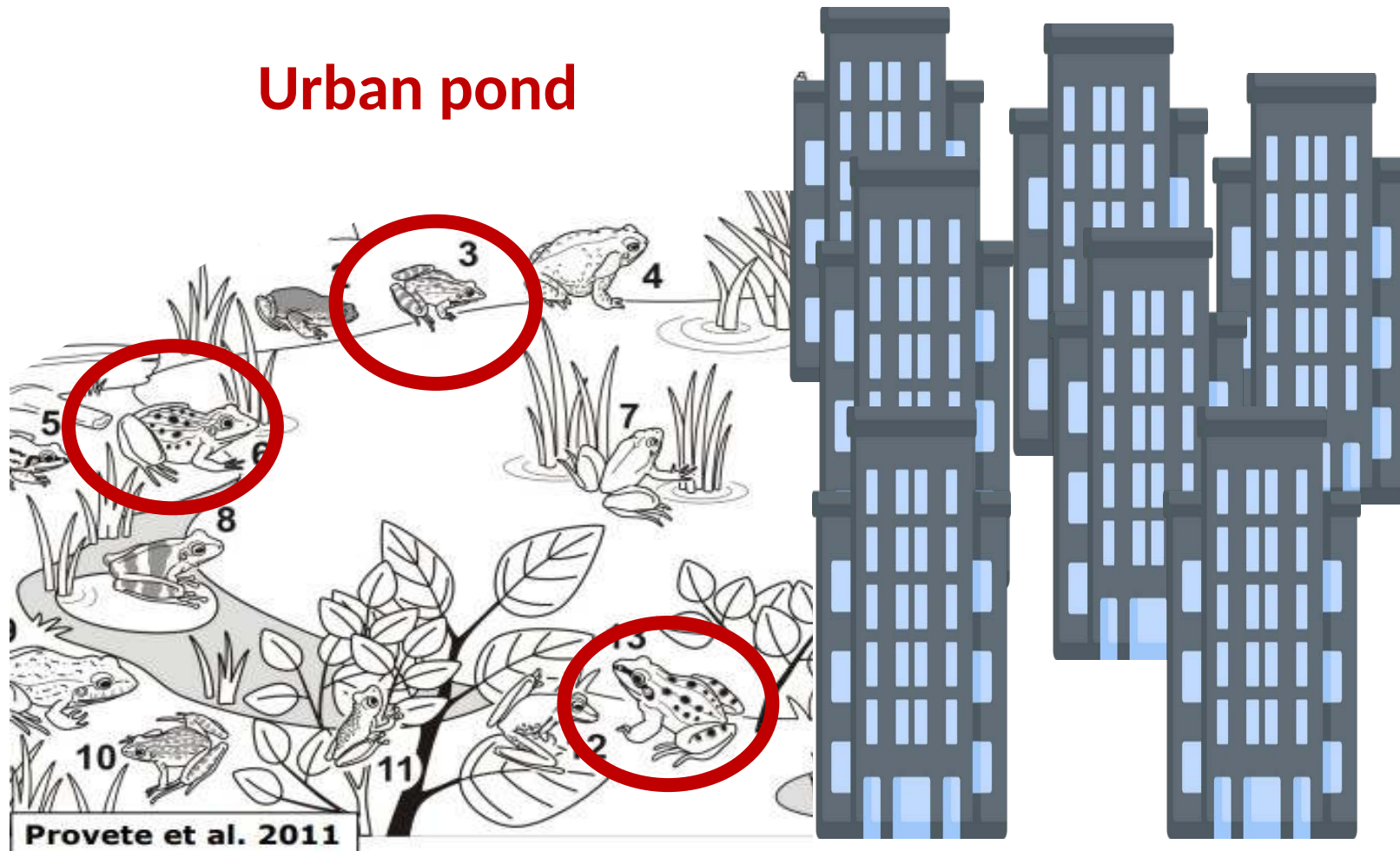
Results



Local environmental variables affect the **interspecific** variation in **head width** ($P = 0.02$), and **landscape** variables affect **interspecific** variation in **leg length** ($P = 0.0005$) in the younger, but not in the older city

Discussão

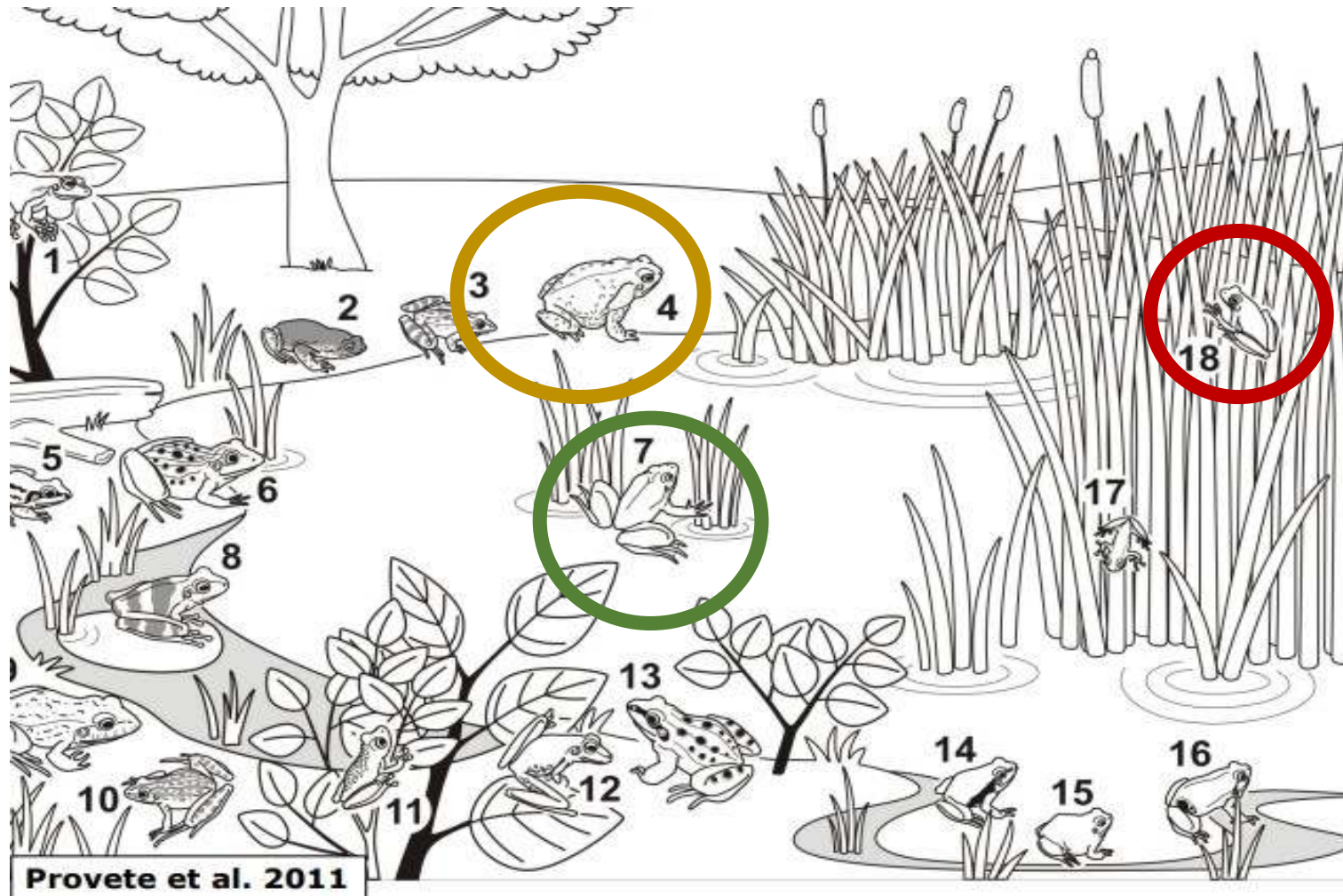
Differences in **habitat availability** drives species turnover, via the requirement niche, affecting species **coexistence**; and heterogeneity



Peters 1986, Chase & Leibold 2003

Differences in habitat availability drives species turnover, via the requirement niche, affecting species coexistence;

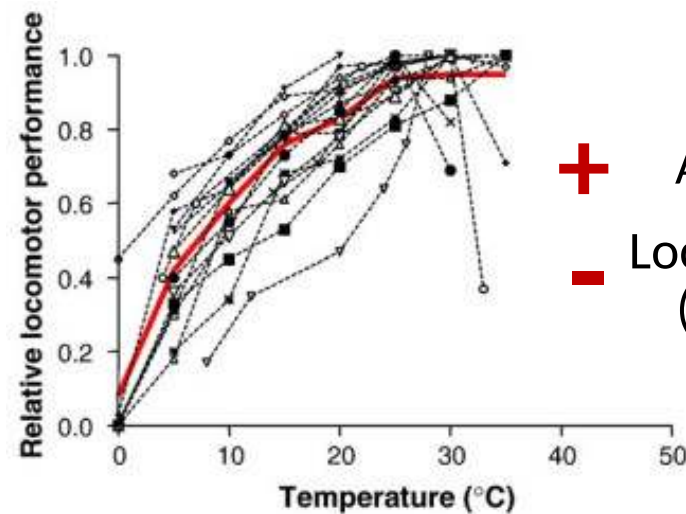
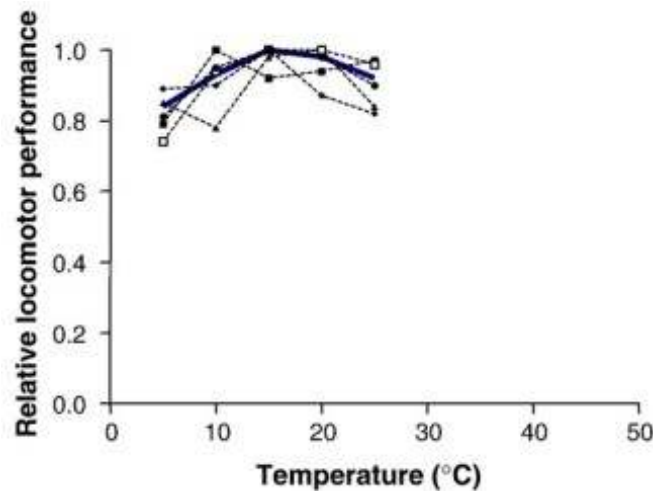
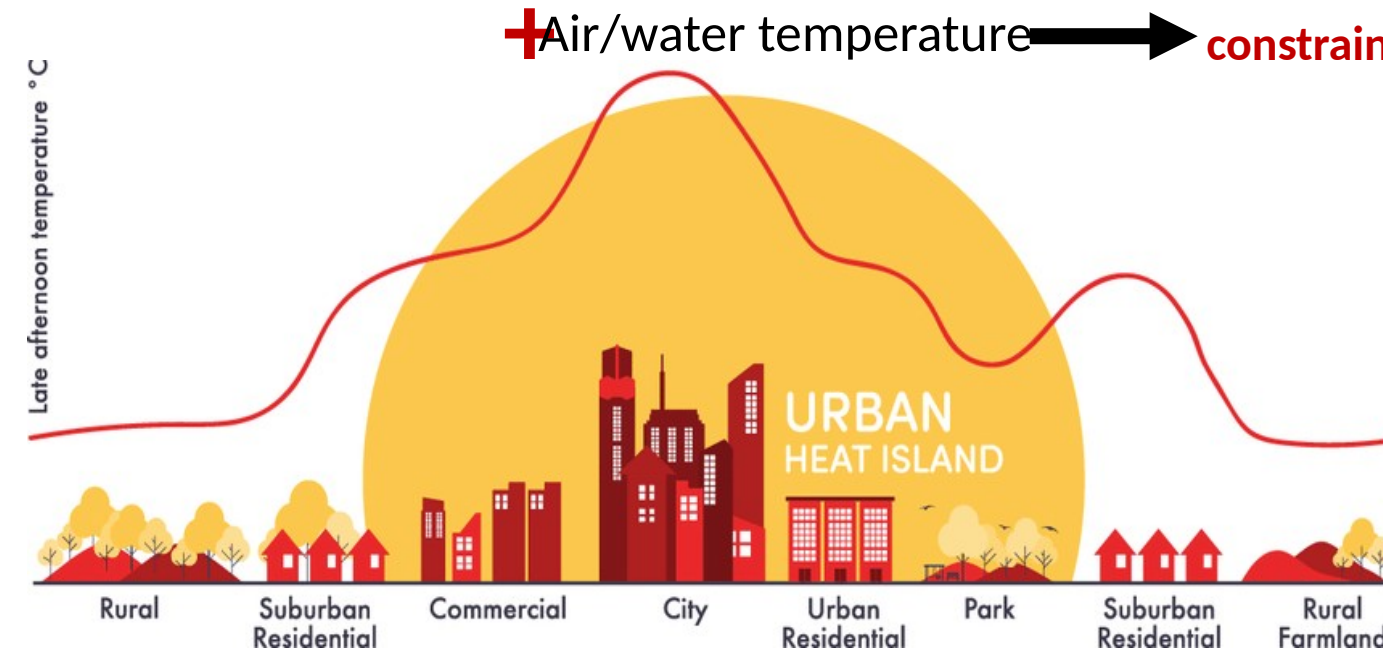
Rural pond



Discussion

+ Air/water temperature → **constrain** the intraspecific variation in relative leg length

Positively related to PC1



+ Air temperature
— Locomotor performance (requirement niche)



Closing remarks

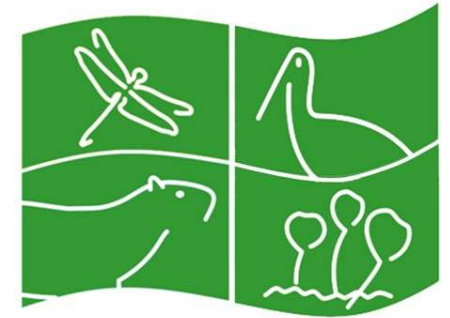
- Due to recent urbanisation, selection pressure is not very high
- Quantitative phenotypic traits provide another perspective on ecology, clear link to functional aspects
- Functional homogenisation (low variability) in urban ecosystems
- Effects of urbanisation on freshwater organisms (e.g., frogs) seem to be indirect, via changes in environmental

Acknowledgements

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